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PGC1α expression using targeted redox-responsive nanogels protects against prostate cancer *in vivo*

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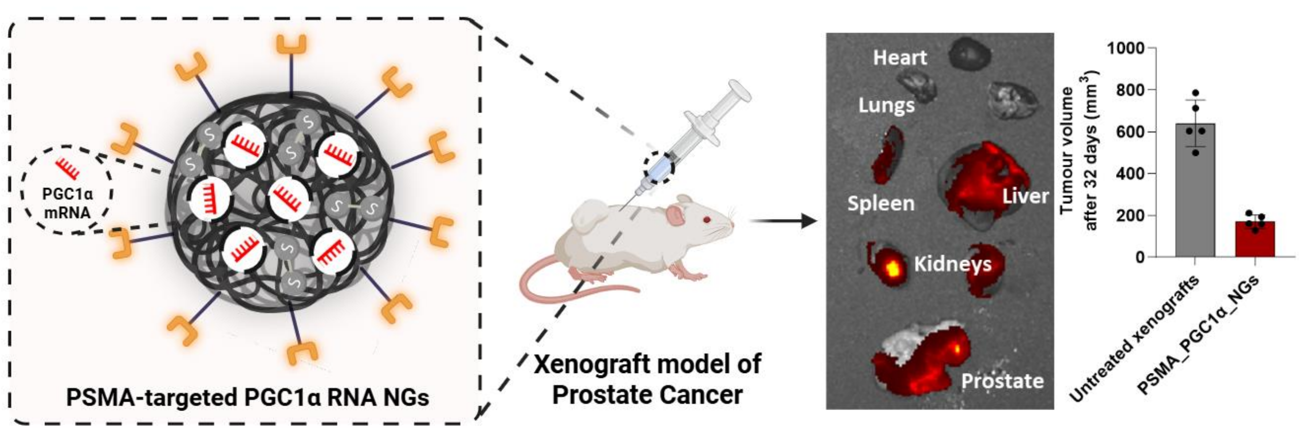

ABSTRACT

Prostate cancer is among the most frequently diagnosed cancers in men in the UK and US. Increasing evidence implicates metabolic dysregulation as a critical driver of disease progression. Central to this process is peroxisome proliferator-activated receptor gamma coactivator 1-alpha (PGC1α) that promotes oxidative metabolism and mitochondrial biogenesis while inhibiting metastatic programs. This work investigated the therapeutic potential of PGC1α overexpression via mRNA delivery. Here, we report a prostate-specific, targeted disulphide-crosslinked nanogel system for intracellular delivery of mRNA encoding the N-terminal isoform of PGC1α (NT-PGC1α). Functionalization of the NGs with a peptide targeting prostate-specific membrane antigen (PSMA) enabled selective delivery of NT-PGC1α mRNA in PCa cells and 3D spheroid models. We confirmed sustained PGC1α expression, and increased mitochondrial protein content, indicative of enhanced mitochondrial biogenesis. These nanogels, which were prepared *in situ* using a nanopolymerization technique, exhibited high mRNA loading capacity, low cytotoxicity, and redox-responsive cargo release, enabling controlled cytosolic delivery following intracellular glutathione-mediated degradation. *In vivo*, systemic administration of the PSMA-targeted NT-PGC1α mRNA-loaded nanogels resulted in tumor-preferential accumulation and significant suppression of xenograft growth (by 73.4% relative to untreated control), with minimal systemic toxicity. This study presents the first example of a prostate-targeted, disulfide-crosslinked nanogel system for mRNA-mediated metabolic reprogramming in prostate cancer, and highlights its promise as a platform for future RNA-based targeted and precision stimuli-responsive cancer therapies.

INTRODUCTION

Prostate cancer (PCa) is now the most commonly diagnosed cancer in men in the UK and US, with rising incidence contributing to a growing global health burden.[1,2] While early-stage disease is often well-managed through surgery, radiotherapy or androgen deprivation therapy, progression to advanced, treatment-resistant forms remains a significant clinical challenge.[3] Increasing evidence implicates metabolic dysregulation as a key driver of PCa progression and therapeutic resistance.[4,5] Indeed, these allow tumors to maintain cell survival, rapid proliferation, resist apoptosis and adapt to environmental stressors such as hypoxia and nutrient deprivation.[3] While oncogenic signalling pathways trigger many of these changes, their full implementation requires coordinated regulation of complex metabolic networks. The transcriptional co-activator peroxisome proliferator-activated receptor gamma co-activator 1 alpha (PGC1α), a master regulator of mitochondrial biogenesis and oxidative metabolism, has emerged as a crucial node in this regulatory landscape.[6]

In the context of PCa, where metabolic flexibility plays a key role in tumor progression and therapy resistance, it is unsurprising that the dysregulation of PGC1α activity can contribute to the altered energetic and biosynthetic demands of malignant cells.[7,8] Given its unique capacity to orchestrate broad transcriptional changes through interactions with different transcription factors, PGC1α represents an attractive therapeutic candidate to oppose oncogenic metabolic programs.

From a clinical standpoint, impaired mitochondrial function and the shift toward glycolytic and stress-adaptive metabolic states are increasingly recognized as hallmarks of aggressive and treatment-resistant PCa.[9] Reduced oxidative phosphorylation capacity has been associated with progression to castration-resistant disease, metastatic potential, and poor patient prognosis.[9,10] Restoration of mitochondrial biogenesis and oxidative metabolism has therefore emerged as a promising strategy to counteract tumor metabolic plasticity, resensitize cancer cells to therapeutic stress, and limit adaptive resistance mechanisms. By promoting mitochondrial integrity, redox balance, and energy homeostasis, reactivation of PGC1α-regulated pathways has the potential to suppress tumor growth while simultaneously enhancing susceptibility to existing treatments, including androgen deprivation and chemotherapy.[7] Previous work by Penfold *et al*., demonstrated that the overexpression of PGC1α in human PCa cells increases ATP levels, decreases the rate of cell proliferation and cell invasiveness.[11] However, the translational exploitation of PGC1α has been limited by the lack of effective delivery strategies capable of modulating its activity in a tumor-selective manner *in vivo*.

A potential therapeutic approach is to use an mRNA-based delivery which offers a transient yet highly controllable approach to restore functional PGC1α expression within PCa cells. However, mRNA is inherently unstable and susceptible to enzymatic degradation *in vivo*, necessitating the development of a suitable delivery platform.[12–14] While lipid nanoparticle (LNP) systems have demonstrated clinical success, including in the delivery of COVID-19 vaccines, concerns remain regarding their dose-limiting toxicity, limited chemical tunability, and minimal spatiotemporal control over mRNA release.[15,16] Therefore, alternative nanocarrier systems capable of improving RNA stability, intracellular release, and tumor selectivity are of growing interest. To this end, disulphide-crosslinked polymeric nanogels (NGs) represent a promising molecularly-designed platform for RNA delivery. NGs are nanoscale hydrogels with high loading capacities, low toxicity, and tunable release properties.[17–21] In previous work, we demonstrated that our crosslinked NGs minimise premature leakage of the RNA payload, while disulphide linkages enable redox-sensitive degradation in the cytosol, where glutathione (GSH) concentrations are significantly higher (1–10 mM) than in the extracellular environment (2–20 μM).[22–25] This redox-responsive behaviour facilitates controlled intracellular release of functional RNA following endocytic uptake. Furthermore, NGs can be functionalized with targeting ligands to enhance selective delivery to specific cell types or tumor markers.[26–28]

As such, this work builds upon our previous *in vitro* nanogel platform[22] and aims to investigate a potential PCa targeting strategy *in vivo*, exploiting the prostate-specific membrane antigen (PSMA) receptor that is highly expressed on the surface of many PCa cells. Indeed, several PSMA-targeted therapies have been developed including radionuclide-labelled small molecule PSMA inhibitors (^{68}Ga-PSMA and ^{177}Lu-PSMA-617), antibody drug conjugates (MLN2704, PSMA-MMAE, MEDI3726) and cellular immunotherapy (PSMA-targeted bispecific T-cell engagers).[29,30] These have been extensively used in clinical settings for PCa imaging and therapy, thereby highlighting the potential of PSMA as an active target.

Compared to bulky monoclonal antibodies, PSMA-targeting peptides could offer a more chemically tractable approach for conjugation to NG surfaces, with minimal alteration to their physicochemical properties.[31] The peptide-NG construct couples a clinically validated targeting epitope (PSMA) to a chemically tractable, redox-responsive mRNA-delivery vehicle, and it is this combination that defines the novelty of the platform. In this study, we present a PSMA-targeted, disulphide-crosslinked NG system for the selective delivery of mRNA encoding for the N-terminal isoform of PGC1α (NT-PGC1α) to PCa cells (Figure 1). Following synthesis and optimization, a lead *in situ* cross-linked and co-polymerized NG was identified that exhibited efficient mRNA encapsulation, high transfection efficiency, and robust redox-responsive mRNA release via glutathione-mediated cleavage. The comprehensive delivery characterisation of the disulphide-crosslinked NG platform - encapsulation and loading efficiencies across multiple mRNA payloads, RNase-protection assays, serum stability, glutathione-triggered release kinetics, intracellular trafficking and endosomal escape - has been reported in detail in our prior work ([22–25]). Targeting specificity was rigorously validated *in vitro* using PSMA-positive (C4-2) and -negative (PC3) cell lines, to confirm preferential internalization of the PSMA-targeted NGs into the target cells, sustained NT-PGC1α expression, and functional metabolic effects. Transcriptomic profiling revealed upregulation of tumor suppressor pathways and suppression of proliferative gene signatures following PGC1α delivery. In addition, the use of 3D spheroid models enabled a more accurate approximation of tumor microarchitecture and nanobio interactions. Finally, *in vivo* biodistribution studies using a fluorescent mRNA cargo, followed by therapeutic evaluation in a PCa xenograft model, confirmed tumour-preferential accumulation of the mRNA payload and significant growth inhibition following systemic administration of NT-PGC1α NGs. Collectively, this work presents the first example of a prostate-targeted, redox-responsive NG platform for mRNA-mediated metabolic reprogramming in PCa. By restoring PGC1α activity, this nanotherapeutic approach offers a strategy for treating metabolically dysregulated cancers and expands the toolkit of RNA delivery systems alongside established lipid-based vectors.

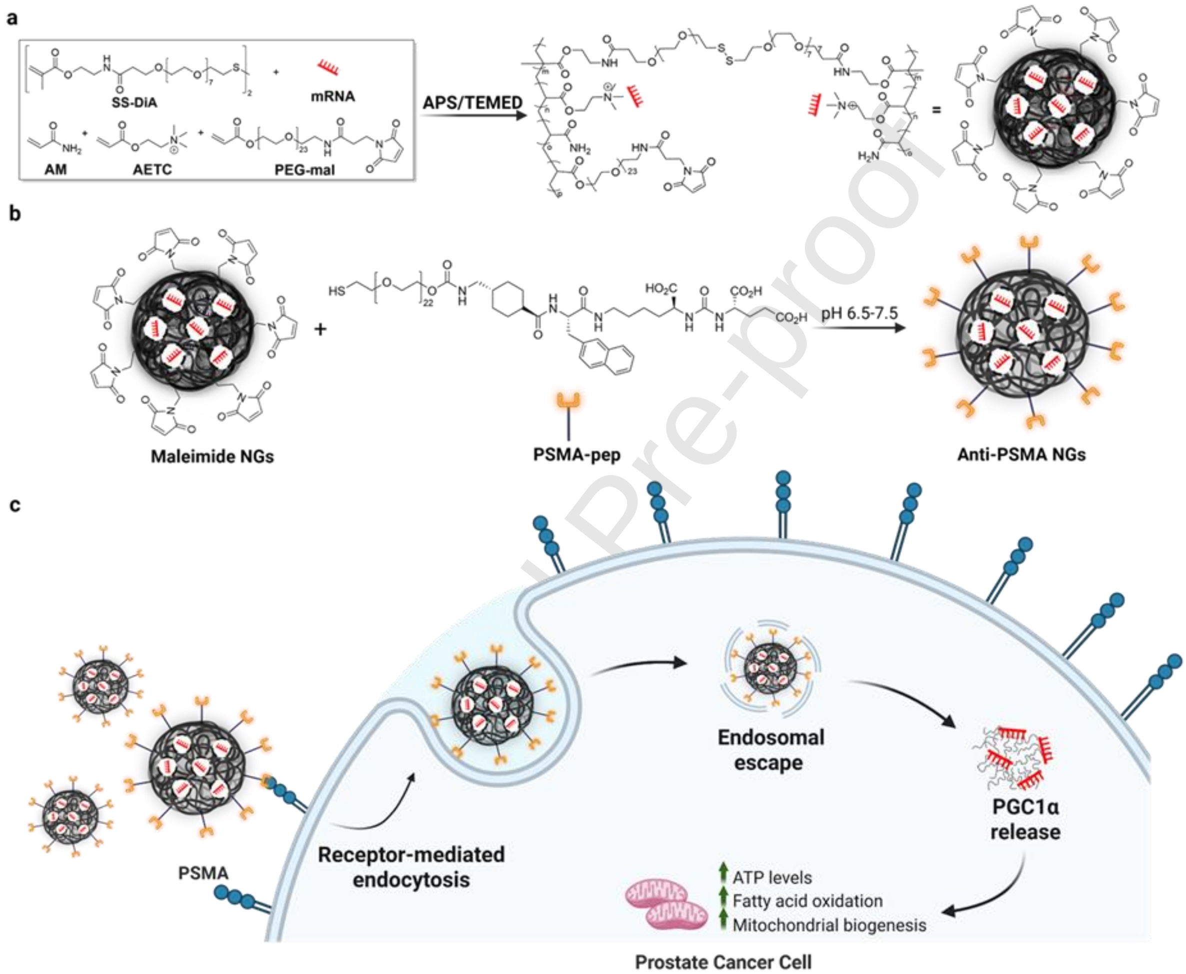

Figure 1: Overall representation of the design, synthesis, and proposed biological function of mRNA-loaded, PSMA-targeted nanogels (NGs). a) Monomer composition and formation of the NG network, incorporating acrylamide (AM), 2-[(acryloyloxy)ethyl]trimethylammonium chloride (AETC) to promote electrostatic templating of the mRNA payload, a disulfide-based crosslinker (SS-DiA) enabling reductive intracellular degradation, and acrylate-PEG-maleimide (PEG-mal) for subsequent ligand functionalization. Free-radical polymerization was triggered by the addition of ammonium persulfate (APS) and N,N,N',N'-tetramethylethylenediamine (TEMED) that allowed simultaneous crosslinking and co-polymerization of the added monomers, resulting in *in situ* encapsulation and stabilization of the mRNA payload. b) Post-polymerization thiol-maleimide conjugation of the PSMA-targeting ligand to PEG-mal–functionalized NGs. c) Proposed mechanism

of cellular interaction in which PSMA binding drives receptor-mediated endocytosis and intracellular delivery of the mRNA cargo.

RESULTS & DISCUSSION

Design and characterization of RNA-loaded NGs targeting the prostate-specific membrane antigen (PSMA).

To achieve targeted delivery into PCa cells, PSMA, was selected as the molecular target due to its abundant and consistent expression in PCa.[3] Small molecule PSMA inhibitors labelled with radionuclides (^{68}Ga-PSMA and ^{177}Lu-PSMA-617) have been widely used for PCa imaging and therapy, demonstrating their high specificity and favourable safety profile.[31,32] The effectiveness of these inhibitors arises from a lysine-urea-glutamate motif which binds to the active binding site with high affinity.[33] ^{177}Lu-PSMA-617 also includes 2-napththyl-L-alanine and tranexamic acid, of which the napthylic amino acid is crucial for improving interactions with the lipophilic binding pocket.[29,33–36] Inspired by this structure, a targeting peptide incorporating the key binding motifs was used. To enable installation of the peptide onto the NG surface via thiol–maleimide click chemistry, which was chosen for its high specificity and yield, a free thiol was introduced at the N-terminus (Figure S1,2). Subsequently, functionalizable RNA-loaded NGs were synthesized by incorporating a bifunctional monomer, acrylate-PEG-maleimide (PEG-mal), during the polymerization process, thereby allowing post-polymerization conjugation of the targeting ligand onto the NG surface via a click chemistry reaction between the free thiol on PSMA-pep and the NG-surface exposed maleimide groups (Figure 1). The maleimide assay was performed to confirm the presence of sufficient unreacted maleimide groups on the nanogel surface for subsequent PEG–peptide conjugation. While some incorporation of maleimide into the polymer network during radical polymerization cannot be excluded, the assay demonstrates that a proportion remains accessible for functionalization. Successful conjugation is further supported by the observed changes in particle size and surface charge following PEG–peptide attachment.

Prior to conjugation with the targeting peptide, the amount of PEG-mal incorporated during NG synthesis was optimized to achieve an appropriate maleimide surface density. The highest PEG-mal mol% yielded the smallest NG sizes (145.3 ± 1.6 nm), lowest PI (0.07 ± 0.03), a favourable zeta-potential (21.8 ± 0.9 mV) and optimal EE% and LE% (Figure S3a-c,e). To assess the presence of residual maleimide groups that remain for surface ligand conjugation post NG synthesis, we measured surface maleimide concentration using a fluorimetric assay.[37] Increasing the PEG-mal mol % led to a corresponding increase in maleimide concentration within the NGs (Figure S3d). The surface density of the maleimide groups ($\frac{\#\, mol}{nm^2}$) was therefore determined, and reflected the trend in increasing maleimide concentration, reiterating the successful conjugation of maleimide moieties onto the NG surface (Figure S3d). Based on these optimization results, 0.1 mol % PEG-mal was selected for subsequent PSMA-labelling studies. Furthermore, to confirm whether the maleimide moieties remained intact for post-polymerisation functionalisation of the NGs, *in situ* ^{1}H NMR monitoring of the polymerisation reaction was carried out. The distinctive singlet at ~6.8 ppm, corresponding to the two H atoms of the maleimide vinyl double bond is clearly present throughout the polymerisation and persists in the final nanogel formulation, while the signals associated with the acrylate vinyl H atoms are consumed over the course of the reaction (Figure S4). This provides spectroscopic evidence of maleimide survival during polymerisation.

Following the optimisation studies, the PSMA targeting peptide (PSMA-pep) was conjugated to the NG surface, via a stable thioether bond between PSMA-pep and the maleimide-coated NGs and ligand density was optimized. As the peptide lacks any other exposed free thiols, this ensures site-specific bioconjugation, without compromising its bioactivity. Notably, increasing PSMA-pep concentration to 0.052 µmol resulted in larger NG size (256.9 ± 4.1 nm) and a higher PI (0.10 ± 0.02) as well as a less positive zeta-potential (8.9 ± 0.9 mV) (Figure 2a,b,e, Figure S5). While changes in zeta potential can reflect alterations in colloidal stability, the values observed here are not sufficiently low to indicate intrinsic instability of the NGs. Instead, the shift in zeta potential with increasing PSMA-pep content is more likely attributable to the net charge characteristics of the peptide itself. The PSMA-pep sequence contains multiple ionizable residues, and its overall charge and isoelectric point within the measurement pH range would influence the surface potential of the PEG-pep–modified NGs. As the cationic character of the NGs originates primarily from AETC, increasing the proportion of PEG-peptide is expected to reduce surface charge density by partially masking or diluting the contribution of AETC, thereby aligning with the observed trend.

In addition to charge effects, incorporating the highest amount of PSMA-pep resulted in an increased in hydrodynamic diameter. The large NG size achieved using the highest amount of PSMA-pep was likely to prevent cellular internalization.[17] As is evident in the TEM images corresponding to 0.052 µmol PSMA-pep, NG morphology seems to be distorted more so than for lower concentrations of the peptide (Figure 2e). After peptide conjugation, the surface PSMA-pep surface coverage was calculated by referring to the difference of absorbance at 280 nm between the targeted and untargeted NGs (Figure 2d). The napthyl alanine moiety provides strong absorbance at 280 nm and was used to determine the approximate measure of peptide surface concentration and density (see experimental section for details). Increasing the PSMA-pep concentration led to a corresponding decrease in maleimide surface concentration (Figure 2d). Notably, 0.052 μmol PSMA-pep did not result in a substantial reduction in NG surface maleimide coverage compared to 0.026 μmol and was therefore excluded from further optimization. 0.026 μmol PSMA-pep provided the most optimal size (202.0 ± 1.0 nm) and polydispersity index (PI) (0.04 ± 0.02). Although the zeta-potential (12.0 ± 0.5 mV) was lower than the desirable range,[17] the increased PSMA-pep surface concentration (0.02 ± 0.001 μmol) and density (19.1 ± 1.9 $(\# \, mol)/(nm)^{\wedge}2$) compensated for this decrease without any adverse effects on NG morphology. As such, this concentration of PSMA-pep was chosen for investigating the *in vitro* targeting efficiency of the labelled NGs.

In parallel, the optimal time frame for the post-coupling conjugation reaction was also investigated. A 12 h timepoint was selected for the conjugation as it provided NGs of desirable sizes (201.3 ± 22.5 nm; PI 0.14 ± 0.04) and consistent zeta-potential (12.0 ± 0.2 mV) (Figure S6).

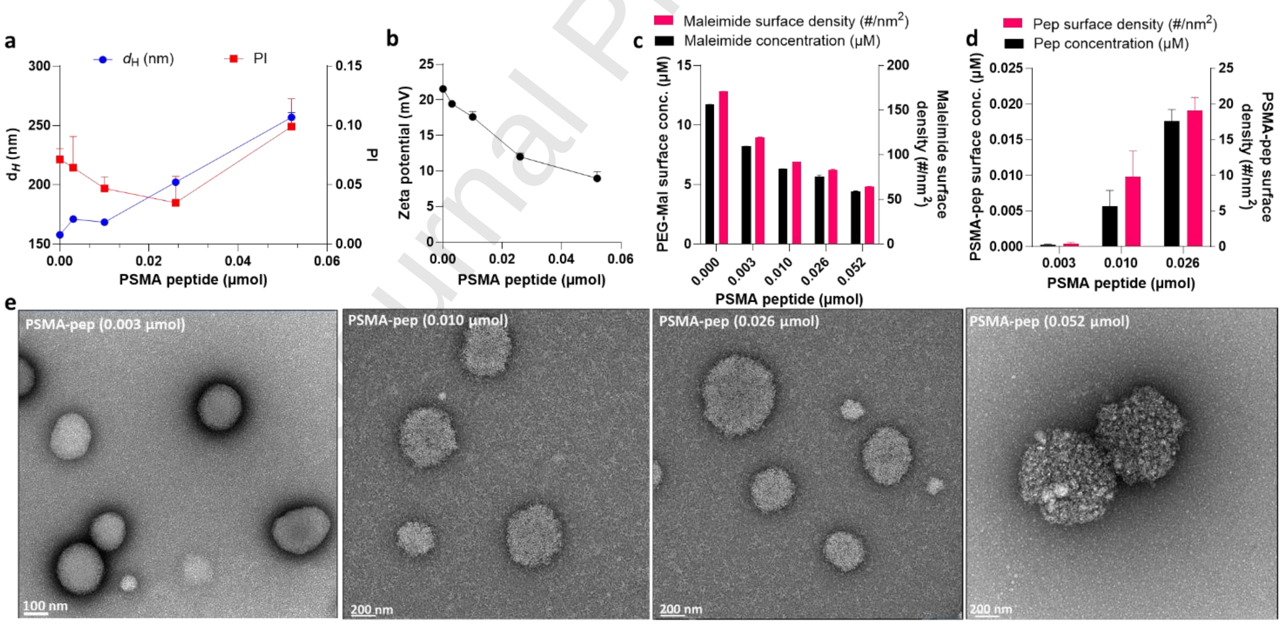

Figure 2: Physicochemical characterization of RNA-loaded PSMA-pep incorporating NGs. Effect of varying PSMA-pep amount on a) NG size, PI, b) zeta-potential as measured by DLS and ELS, respectively, c) maleimide surface concentration and coverage, and d) calculated values for NG surface concentration and density of PSMA-pep. In all cases, data are presented as mean ± standard deviation for n = 3. e) Representative negative stain TEMs of NGs at 45,000 x and 200kV. Scale bar at 200 nm.

As a further assessment of PSMA conjugation on NG surface, single particle automated Raman trapping analysis (SPARTA) was used (Figure S7).[38] SPARTA, offers simultaneous analysis of size and composition at a single particle resolution. By integrating optical trapping with Raman acquisition, it provides a molecular fingerprint of each particle, such that subtle surface modifications can be detected.[38] Compared to IR spectroscopy, Raman analysis via SPARTA offers distinct advantages, particularly for nanoparticles dispersed in aqueous media, since Raman signals are minimally affected by water absorption. Additionally, Raman's sensitivity to non-polar bonds and certain functional groups enhances its suitability for detecting surface chemistry modifications such as PSMA conjugation. Furthermore, SPARTA analyses each particle individually, it also enables quantitative assessment of heterogeneity in surface modification across a population, which would be masked in bulk measurements.

As the final NG construct is a molecularly structured, multi-monomer-based nanoparticle system, NGs were synthesized using each monomer in isolation as well as in various combinations to serve as controls. This approach enabled direct comparison and identification of the individual monomer contributions within the final formulation. Single monomer systems included AETC only NG (AETC_NG), acrylate-PEG-maleimide only NG (MAL_NG), and disulphide-crosslinker only NG (PEGSS_NG) (Figure S7a-c). Notably, NGs could not be synthesized with acrylamide alone, likely because this leads to a free formed linear polymer rather than a spherical NG.

For the single monomer systems, common regions of interest pertain to *v*(C-N) from the C–C–O backbone at 1450 cm^{-1} and rocking bending modes from NH_2 moieties (ɣ(NH_2)) at ~1295 cm^{-1} as well as CH_2 moieties (ɣ(CH_2)) at ~ 1060 cm^{-1} (Figure S7a-c). Notably, maleimide-incorporating NGs demonstrate a distinct peak in the region ~1600-1640 cm^{-1} (Figure S7b). This peak only appears in the spectra of the NG synthesized with all monomers combined, suggesting a contribution from carbonyl-containing functional groups, consistent with the presence of maleimide-derived structures (Figure S7f,g). The spectrum for the complete PSMA-functionalised NG, which includes all monomers, is presented in Figure S7, revealing the cumulative vibrational features. In the NG formulation containing all monomers, this broad maleimide peak was notably absent, consistent with changes in surface composition following PSMA peptide conjugation. Instead, a pronounced amide I band at ~1606 cm^{-1} appeared, consistent with amide bonds in the peptide backbone, consistent with successful PSMA conjugation. Furthermore, smaller peaks in the regions ~800-880 cm^{-1} (PO^{2-} symmetric stretch), ~1090 cm^{-1} (PO^{2-} asymmetric stretch), ~1480-1500 cm^{-1}, indicating RNA entrapment within the NGs.[39,40] To our knowledge, this study is the first to develop a PSMA-targeted polymeric nanoparticle system for mRNA delivery.

***In vitro* specificity assessment of PSMA-targeted NGs.**

After targeted nanogel synthesis and characterization, we determined the cellular targeting ability of the PSMA_NGs. To establish a suitable model system for evaluating relative uptake efficiency, three PCa cell lines as well as cancer-associated fibroblasts (CAFs) were screened for their PSMA expression levels. As reported previously, C4-2 cells demonstrate high levels of PSMA expression, whereas PC3 demonstrate negligible expression (Figure 3a).[41,42] Since these cell lines demonstrated the greatest differential PSMA expression among those tested, they were chosen for subsequent cellular uptake investigations.

We used fluorescently labelled NGs (developed in previous work[22]) that were synthesized to be targeted (PSMA_NG) and untargeted (Ut_NG) in this work. It was hypothesized that PSMA_NG uptake would be reduced in both PSMA-low cells and in cells pre-treated with an anti-PSMA antibody, the latter expected to competitively block receptor binding and thereby limit internalization. Using confocal microscopy in tandem with flow cytometry, PSMA_NG uptake was lower in PC3 cells (microscopy: 9.2 ± 4.0 %, cytometry: 29.6 ± 0.5 %), compared to C4-2 cells (microscopy: 61.5 ± 5.3 %, cytometry: 76.9 ± 3.5 %) (Figure 3b-d, Figure S8). Notably, the uptake of (untargeted NGs) Ut_NGs in PC3 cells was comparatively unaffected following incubation with anti-PSMA antibodies (microscopy: 39.0 ± 7.7 %, cytometry: 31.6 ± 2.3 %) (Figure 3c). Strikingly, C4-2 cells pre-treated with the anti-PSMA antibody before PSMA_NGs incubation showed a 96.6 % reduction in fluorescence signal in microscopy images and a 58.2 % decrease in FITC-positive signal by flow cytometry (Figure 3b). In contrast, the uptake of Ut_NGs following anti-PSMA treatment remained largely unaffected in both cell lines. As further confirmation of the specificity of the PSMA_NGs for the PSMA receptor, immunofluorescence studies were performed to investigate the colocalization of the PSMA signal (red) with the fluorescent NG (green) signal using confocal microscopy. A higher Pearson's correlation coefficient was observed in PSMA_NG-treated cells (r=0.65) as opposed to those treated with Ut_NGs (r=0.29) (Figure S9).

The mode of cellular internalization was studied by investigating NG uptake in the presence of inhibitors comprising all known uptake pathways.[43] Although C4-2 and PC3 cell lines internalise nanoparticles through endocytosis, in the case of C4-2 cells, it is likely that the PSMA peptide interacts with the PSMA receptor to undergo receptor-mediated endocytosis via clathrin-coated vesicles.[44–46] Indeed, a threefold increase in PSMA internalization was observed following the binding of specific ligands.[47,48] Treatment with hyperosmolar sucrose and NaN_3/2-deoxy-D-glucose inhibited the uptake of both NG types, suggesting clathrin-mediated endocytosis remains the predominant pathway (Figure S10).

Next, C4-2 spheroids were cultured to investigate the NG trafficking and penetration in a three-dimensional system (Figure 3f,g). Z-stack images were acquired and analyzed using Fiji software, where fluorescence intensity was quantified as integrated density (total fluorescence signal within a defined area) across multiple slices, allowing for a more accurate assessment of NG distribution throughout the entire spheroid volume. Interestingly, even within a 24h period, a marked increase in the uptake of PSMA_NGs was observed compared to the Ut_NGs, underscoring both the efficiency of targeted delivery and the importance of employing complementary imaging techniques to confirm results (Figure 3d,e). Given the comparatively larger size of these NGs (~200nm), these results are particularly encouraging, as previous studies have noted a distinct nanoparticle size constraint on organoid penetration efficiency. This highlights a unique advantage conferred by the likely PSMA-mediated internalization mechanism of the PSMA_NGs.[17,49]

Given the widespread use of lipid nanoparticle (LNP) platforms for RNA delivery, we next sought to contextualize the physicochemical properties and cellular uptake behavior of the PSMA_NGs relative to a representative LNP formulation (Figure S11). Comparative analysis of particle size, morphology, and expression profile provides an informative framework for evaluating platform-level differences. Accordingly, PSMA_NGs and LNPs were compared in terms of hydrodynamic diameter (DLS), morphology (TEM), and cellular uptake assessed by confocal microscopy, where the NGs demonstrate smaller hydrodynamic diameters and a different uptake profile to the representative LNP formulation tested under matched *in vitro* conditions (Figure S11). We present this comparison as contextual benchmarking as opposed to a functional head-to-head efficacy comparison.

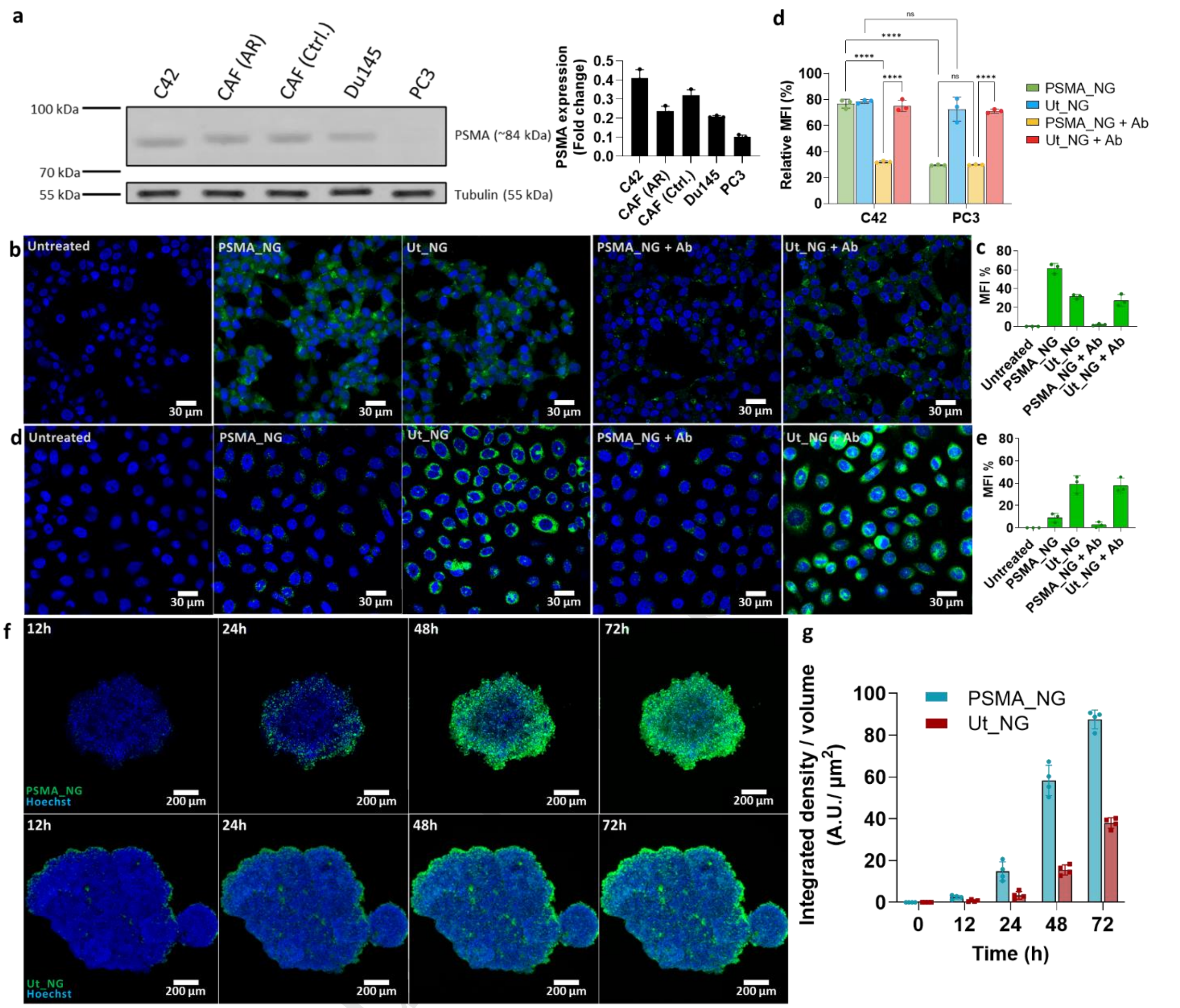

Figure 3: Investigating cellular uptake efficiency of targeted and untargeted NGs. a) Western blot analysis of PSMA expression across PCa cell lines. Cell lysates from C4-2, CAF (androgen receptor positive), CAF (androgen receptor negative), Du145 and PC3 cells were probed for PSMA. Tubulin served as a loading control. Densitometric quantification of PSMA signal intensity, normalized to tubulin, is shown on the right. b) Confocal microscopy images of C4-2 showing the uptake of PSMA_NGs or Ut_NGs with or without PSMA antibody pre-treatment. Scale bar at 30 µm. c) Quantification of mean fluorescence intensity (MFI) from multiple fields of view is shown in the bar graph on the right. d) Confocal microscopy images of PC3 showing the uptake of PSMA_NGs or Ut_NGs with or without PSMA antibody pre-treatment. Scale bar at 30 µm. Data are presented as mean ± SD for n = 5, and statistical significance was determined using a 2-way ANOVA. ****$p < 0.0001$, ns – not significant. e) Quantification of mean fluorescence intensity (MFI) from multiple fields of view is shown in the bar graph on the right. f) Uptake of fluorescent NGs within 72h shown by maximum intensity projections of confocal microscopy images (10 x, slice length: 10 µm) of live spheroids embedded in agarose. Scale bar at 200 µm. g) Quantification of NG uptake by integrated

fluorescence density (sum of pixel intensities across all z-cross sections) normalized to spheroid volume. Data represent mean ± SD from n = 3.

Monitoring *in vivo* biodistribution and targeting efficiency assessment in C57BL/6 mice.

The *in vivo* biodistribution and targeting capacity of the PSMA_NGs was examined next. For biodistribution and initial efficacy studies of the nanogels, the mRNA encoding for mCardinal (ex/em: 604/659nm) was encapsulated and administered at a dose of 0.25mg/kg (polymer dose: 15mg/mL; 60mg/kg), based on dose ranges reported in previous studies using nanoparticles.[50–52] Following intravenous tail-vein injection, the radiant efficiency of the main organs measured *ex vivo,* was determined using the *in vivo* imaging system (IVIS spectrum) at 24h from injection (Figure 4a). As shown in Figure 4b,c, the distribution of both NG types was evident in the liver, spleen and kidneys after 24h, as expected for nanoparticles that are injected intravenously.[53] Furthermore, the results from *ex vivo* imaging were supported by western blot analysis and RT-qPCR analyses, which showed increased protein expression, and mRNA levels were observed in the same organs as before (Figure 4d-f). The NGs are presumed to reach the liver via the hepatic portal vein, and are then eliminated via the hepatobiliary route, which could offer an explanation for the relatively high level of expression in the kidneys.[53] Further, NG presence in the spleen could be explained by uptake by the mononuclear phagocytic system (MPS).[53]

Notably, the targeted NGs (PSMA_mC_NGs) exhibited increased expression in the prostate compared to controls, thereby exemplifying the targeting capability of our platform (Figure 4b,c). This result was echoed in the mRNA levels and protein expression analysis, where a significant increase in mCardinal expression was noted for the targeted formulation as compared to the untargeted NGs in the prostate (Figure 4d,e,f). Despite the non-specific expression in other organs, this work represents the first example of a multi-functional NG - specifically targeting the prostate, marking a crucial step towards the development of a more efficient and targeted delivery platform for PCa therapy.

The time-dependent expression profile of the targeted NGs was then assessed by harvesting organs of interest at 6, 24 and 48h (Figure S12a,b). As mRNA levels decline, protein expression also decreases, albeit with a delay, reflecting the time required for translation and protein turnover (Figure S12c-e). While protein expression was already evident at 24h, as seen previously, it is notably sustained for up to 48h post-administration which is similar to the expression dynamics observed for typical lipid nanoparticle (LNP)-based formulations at similar doses.[15]

A key point of divergence from the LNPs with our NGs lies in the delayed RNA release observed with the disulphide-based NGs. Whereas LNPs often mediate *in vivo* transfection within 6h, likely due to a burst release profile, the disulphide NGs demonstrate a more gradual, biologically triggered release.[54] This feature provides greater control over RNA release kinetics, offering opportunities for fine-tuning delivery timing to suit specific therapeutic or controlled release applications.

Two modes of administration were investigated to explore the potential for repeated dosing of the animals with the functional PGC1α mRNA. We conducted a study in a xenograft model of PCa in NSG mice to assess the difference in mCardinal expression in tumors between mice treated with mC_NGs or PSMA_mC_NGs and to evaluate whether the mode of administration (subcutaneous vs intravenous injection) influenced the expression profile and biodistribution of the NGs (Figure S13a). For both modes of administration, 24h post-administration, PSMA_mC_NGs demonstrated uptake in the prostate and tumor, as well as some non-specific accumulation in the liver, spleen and kidneys, as before (Figure S13c-f). Quantitative tumor-to-organ expression ratios revealed tumor-preferential accumulation relative to spleen and prostate, while liver and kidney exposure remained comparable to or greater than tumor levels. Untargeted mC_NGs also exhibited tumor uptake, most likely driven by the enhanced permeation and retention (EPR) effect, which is primarily governed by the leaky vasculature in the tumor tissue that allows nanoparticles to extravasate from circulation (Figure S13c-f).[55] Comparison of tumor-to-organ ratios between targeted and untargeted NGs indicated that PSMA functionalisation redistributes nanoparticle accumulation toward the prostate–tumor axis relative to untargeted controls, although it does not fully abrogate off-target uptake. While tumor uptake was comparable between both intravenous and subcutaneous routes, a slight reduction in uptake was observed in the liver, spleen and kidneys for subcutaneous injections as compared to IV administration, as shown in the western blot analysis (Figure S13g,h). Notably, in the case of the targeted NGs, an overall increase in accumulation in tumor tissue was confirmed by immunofluorescence analysis on frozen tissue sections (Figure S13i,j). Collectively, these findings indicate that the route of administration does not compromise *in vivo* transfection efficiency of the targeted NGs - representing a key advantage for our targeted NG. However, the persistence of hepatic and splenic accumulation highlights a constrained therapeutic index, consistent with known reticuloendothelial clearance of nanoparticle systems, and underscores the need for further optimisation to mitigate off-target exposure in translational settings.

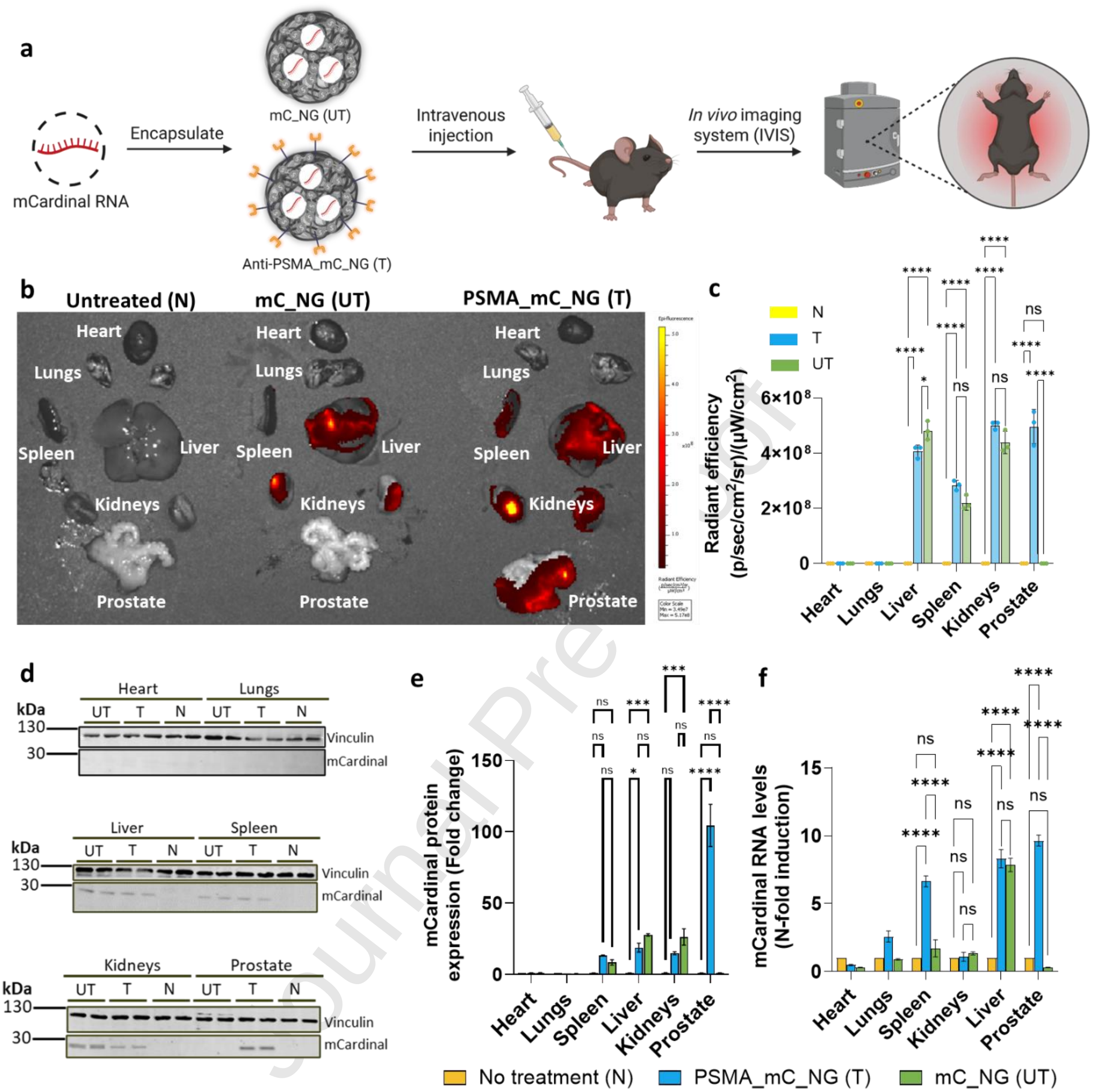

Figure 4: Assessment of *in vivo* biodistribution of NGs in C57BL/6 mice. a) Schematic presentation of the experimental set up where mCardinal mRNA, encoding for a far-red fluorescent protein mCardinal, was encapsulated inside the targeted (PSMA_mC_NG; Targeted (T)) and untargeted (mC_NG; Untargeted (UT)) NGs, that were injected intravenously and the expression monitored and compared to control (C) mice (treated with saline) using the *In vivo* Imaging System (IVIS) 24h post-administration. b) *Ex vivo* fluorescence imaging and c) radiant efficiency quantification of main organs extracted from the mice at 24h after injection of NGs (n = 3). d) mCardinal protein expression in major organs determined by western blotting and e) associated quantification. f) Quantification of mCardinal mRNA expression using RT-qPCR. Expression was normalized to GAPDH expression and shown as fold change relative to saline-treated mice. In all cases, data are presented as mean ± SD from n = 3,

and statistical significance was determined using a 2-way ANOVA. ****p < 0.0001, ***p < 0.005, **p < 0.01, ns – not significant.

Validating PGC1α & NT-PGC1α mRNA overexpression and downstream effects on mitochondrial biogenesis

Owing to PGC1α's beneficial role in PCa suppression[7], the overexpression of the N-terminal truncated version of PGC1α was investigated in PCa cells via NT-PGC1α mRNA-loaded NGs (NT-PGC1α_NG) (Figure 5a). In humans, PGC1α is encoded by the *PPARGC1A* gene, and this gene undergoes extensive alternative splicing.[56] Splicing between exons 6 and 7 of the *PPARGC1A* gene generates a transcript encoding a truncated, N-terminal isoform of PGC1α (NT-PGC1α), that comprises of 270 AAs (Figure 5b).[56,57] NT-PGC1α presents a persistent transcriptional coactivator comprising of the transcription activator and nuclear receptor interaction domains of full length PGC1α. However, it is no longer subject to phosphorylation-mediated turnover, thereby resulting in the generation of a PGC1α variant that has constitutive transcriptional activity with an extended protein half-life.[56,57] Therefore, the transcript for the spliced version of PGC1α was synthesized and entrapped within the NGs (NT-PGC1α_NG).

The NGs resulted in increased NT-PGC1α mRNA and protein expression when compared to the commercial reagents (Figure 5c,f). This increased efficiency was mirrored in two other PCa cell lines tested, PC3 and Du145 (Figure 5d-f). In addition, delivery via NT-PGC1α_NG resulted in a substantial upregulation of the protein (Figure S14a,b). Furthermore, immunofluorescence staining was used to confirm nuclear localization of NT-PGC1α (Figure 5h). Consistent with the western blot results, NT-PGC1α expression was highest in NG-treated cells and showed a greater level of nuclear localization (r=0.747), relative to the other conditions.

Having confirmed overexpression using the NGs in C4-2 cells, the downstream effects of NT-PGC1α were investigated to determine the functional efficacy of the payload. The impact of NT-PGC1α overexpression on mitochondrial abundance and activity was assessed. First, expression of mitochondrial electron transport chain (ETC) proteins in C4-2 cells treated with NT-PGC1α_NG was determined (Figure S14c,d).[6] The increase in mitochondrial number was more pronounced in cells treated with NT-PGC1α_NG compared to those treated with BI9774, indicating that direct delivery of NT-PGC1α induces a stronger mitochondrial response (Figure S14c,d).

To further assess the effects of NT-PGC1α on mitochondrial function cells were stained with MitoTracker™ Green (MTG) and TMRM. MTG, serves as a marker for mitochondrial abundance, whereas TMRM accumulation is sensitive to changes in mitochondrial membrane potential.[58] Flow cytometry was used to generate bivariate dot plots of TMRM versus MTG staining intensity (Figure S14c). These plots revealed distinct shifts in the populations of cells treated with NT-PGC1α_NG toward higher intensities of both dyes, suggesting increases in both mitochondrial content and membrane potential compared to the other conditions. Quantitative analysis of the mean fluorescence intensity (MFI) for MTG (Figure S14d) confirmed a significant increase in mitochondrial abundance in NG-treated cells compared to controls. Similarly, TMRM MFI values (Figure S14e) indicated enhanced mitochondrial membrane potential, with the NT-PGC1α_NG group showing the highest signal among all conditions. To integrate both abundance and functional capacity, the ratio of MTG to TMRM MFI was calculated (Figure S14f). This ratio further confirmed that NG-mediated delivery of NT-PGC1α resulted in the most pronounced enhancement in mitochondrial biogenesis and activity. Mitochondrial function was further assessed by measuring oxygen consumption rate (OCR) (Figure S15a). NT-PGC1α_NG treatment markedly enhanced mitochondrial respiration, with elevated basal and maximal respiration, as well as spare respiratory capacity (a measure of mitochondrial reserve function), indicating greater mitochondrial fitness compared to other groups. Extracellular acidification rate (ECAR) profiles were largely unchanged across conditions, though a modest reduction in glycolysis was noted in NT-PGC1α_NG-treated cells, consistent with a metabolic shift towards oxidative phosphorylation (Figure S15b).

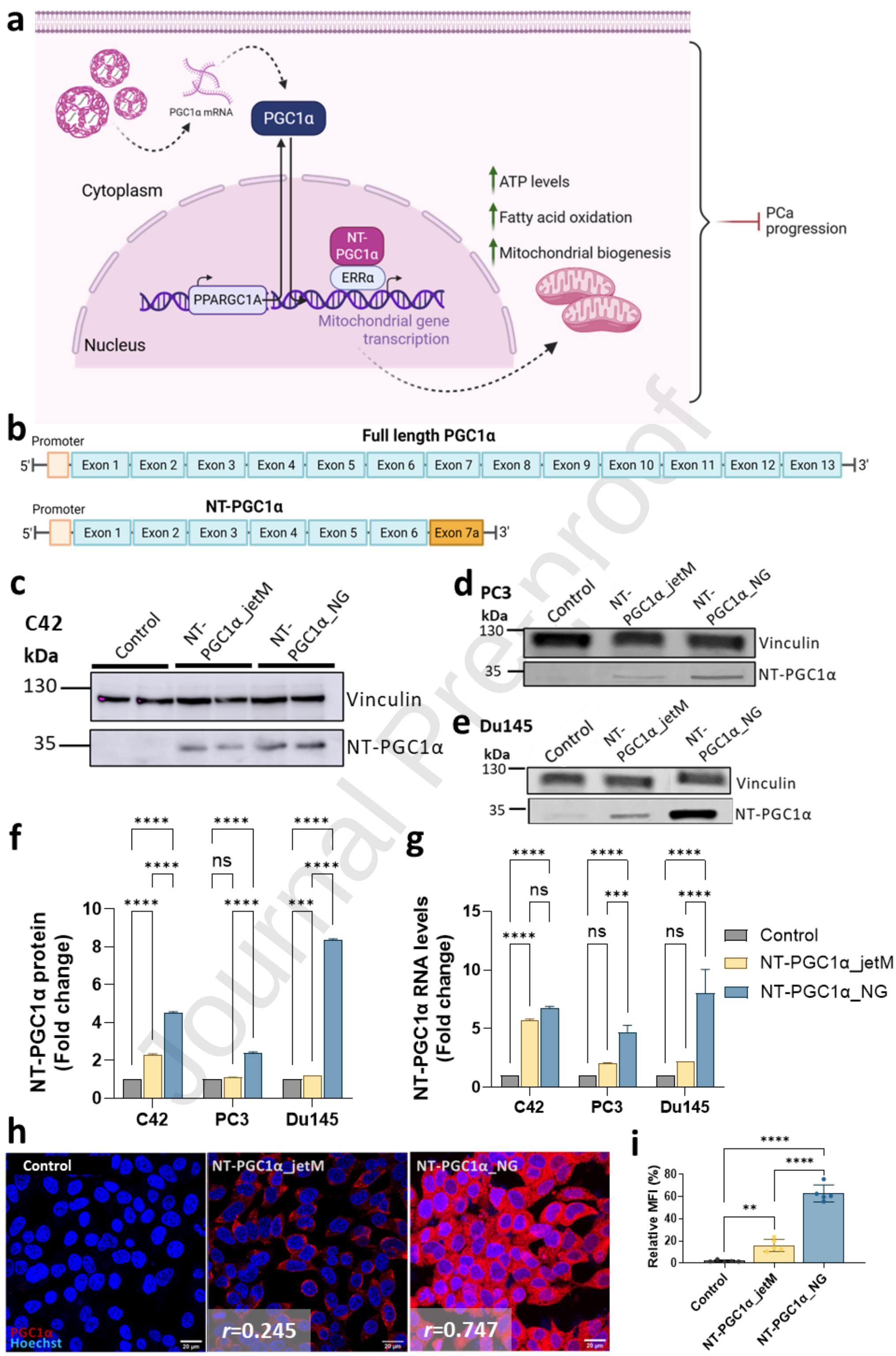

Figure 5: Investigating effect of NT-PGC1α overexpression in C4-2 cells. Schematic presentation a) of the way in which delivery of NT-PGC1α mRNA within NGs leads to overexpression of NT-PGC1α, a key regulator of mitochondrial biogenesis and oxidative metabolism – blocking further PCa cell growth. Elevated PGC1α promotes mitochondrial biogenesis, fatty acid oxidation, and ATP

production, contributing to a metabolic shift away from lipid accumulation; b) of the two predominant splice isoforms of *PPARGC1A*. c) C4-2, d) PC3 and e) Du-145 cells were treated with increasing concentrations of NT-PGC1α_NG, lysed and blotted for NT-PGC1α expression, as detected by anti-NT-PGC1α mAb (sc-5816). f) Quantification of the normalized protein expression is shown (n = 3). g) Comparison of NT-PGC1α mRNA levels following NT-PGC1α_NG administration, as quantified by RT-qPCR. RNA levels normalized to GAPDH expression and shown as fold change relative to untreated cells. h) Maximum projection immunofluorescence images of C4-2 cells either left untreated or treated with BI9774, NT-PGC1α_NG and NT-PGC1α_NG + BI9774 for 48h, fixed and stained with 1 μg/mL Hoechst 33342. Cells were treated with rabbit anti-NT-PGC1α mAb (sc-5816, 1:100) and then treated with Goat anti-Rabbit IgG (H+L) Alexa Fluor^{TM} 568 secondary antibody (1:500). Quantification of the red fluorescence (MFI) is shown on the left for all conditions (n = 5). All images were taken with 63 x oil immersion objective. Scale bar at 20 μm. Co-localization analysis conducted using JACoP revealed Pearson's correlation coefficient values as labelled on the relevant images. In all cases, data are presented as mean ± SD from n = 3, and statistical significance was determined using an ordinary one-way ANOVA with Dunnett's multiple comparisons test, where RNA/protein levels of treated cells were compared against those of untreated cells: ****p < 0.0001, ***p < 0.0002, **p < 0.0011, *p < 0.0322, ns – not significiant.

Previously, Torrano *et al.,* examined the role of PGC1α in suppressing PCa metastasis.[59] In this study, gene expression profiling showed that PGC1α significantly modulates both gene expression and metabolite levels, particularly those associated with mitochondrial catabolic pathways and oxidative processes, such as fatty acid β-oxidation. Collectively, these findings suggested that in PCa, PGC1α functions as a key metabolic regulator, tipping the balance between a catabolic, tumor-suppressive state (characterized by high PGC1α expression) and an anabolic, tumor-promoting state (associated with low PGC1α expression). In a subsequent study, Penfold et al., reported the transient expression of PGC1α decreased cell proliferation.[11] In the current study, C4-2 cells treated with NT-PGC1α_NGs decreased cell proliferation considerably (Figure S16a).

To gain an improved mechanistic understanding of the transcriptome influenced by NT-PGC1α_NG treatment, C4-2b cells were treated in parallel with NT-PGC1α-loaded NGs and empty NGs at matched polymer dose every 24 h for three days and subjected to RNA sequencing analysis, so that the differentially expressed gene set reflects cargo-specific (NT-PGC1α-driven) effects rather than the nanogel vehicle itself. 522 genes were significantly upregulated and 1311 genes were significantly downregulated in NT-PGC1α_NG-treated cells relative to the empty-NG control (fold change > 1.5); the empty-NG arm showed no significant cell-cycle or AR-axis transcriptional signature relative to untreated cells in the same dataset. As expected, the expression of *Ppargc1a,* the gene encoding PGC1α was significantly increased (Figure S16b). Comparison against the Human PGC1α Biocarta dataset was used as a readout of NT-PGC1α activity (Figure S16c). Transcriptomic profiling and gene sequence enrichment analysis reveal a coordinated suppression of proliferative pathways as well as the concurrent activation of potential metabolic stress adaptation mechanisms (Figure S16d,e). Indeed, NT-PGC1α overexpression downregulated several genes involved in cell cycle regulation, including *CCND1, CCNE1/2, CCNA1/2, CCNB2/3, CDCA4/7/8, and CDC25A/C* [60–66], as well as mitotic regulators including, *PLK1, BUB1B and TOP2A* (Figure S16b,d).[67–69] This transcriptional program likely halts both G1/S and G2/M transitions, reinforced by the upregulation of CDKN1A (p21), CDKN1B (p27), and TP53 (Figure S16b,d).[70,71]

Crucially, these changes were in parallel to the suppression of a panel of oncogenic drivers, including *MYC (482, 483), KRAS (484), AURKA (483)* and *CTNNB1 (485)*, as well as the androgen receptor (AR) signaling axis, including AR itself, as well as associated factors, such as ARLNC1 (an AR-regulated long non-coding RNA) and NKX3-1 (a prostate-specific tumour suppressor that is paradoxically AR-regulated).[72,73] Given the central role of AR signalling in PCa pathogenesis and progression, this repression is particularly significant. AR drives the transcription of genes involved in cell cycle progression, metabolism, and survival, and its activity is tightly linked to both the development of PCa and resistance to androgen deprivation therapy.[74] The observed downregulation of AR and its co-factors suggests that NT-PGC1α may disrupt the androgen-dependent transcriptional program that prostate tumors rely on for sustained growth.

GSEA of upregulated transcripts reveals activation of processes associated with cellular homeostasis and metabolic adaptation (ribosome biogenesis and rRNA processing, RNA splicing and transcription elongation, autophagy and mitochondrial autophagy, protein–RNA complex assembly and endosomal transport) (Figure S16b,e). These point towards a reprogramming of proteostasis and organelle quality control. Key metabolic regulators such as *ACACB, MPC1, and GOT1* are also upregulated, consistent with NT-PGC1α's role in promoting mitochondrial oxidative metabolism and fatty acid oxidation. *CDKN1A* and *CDKN1B* are also upregulated consistent with the inhibition of cell cycle.[75] Overall, the overexpression of NT-PGC1α in PCa cells leads to a transcriptional landscape characteristic of a tumour-suppressive state.

Validating effects of NG-mediated NT-PGC1α overexpression on PCa progression in a xenograft model of PCa

Driven by the promising results from the *in vitro* and *in vivo* experiments, a xenograft model of PCa was developed to assess the effect of NT-PGC1α treatment on tumor progression (Figure 6a). Immunecompromised mice were inoculated with C4-2b cells and tumors allowed to grow to a predetermined size (50 mm^3) before randomizing the mice and treating with the following conditions:

1. Untreated control: to assess baseline tumor progression
2. Untargeted NT-PGC1α_NG (Ut-NT-PGC1α_NG): to assess the therapeutic efficacy offered by targeting PSMA
3. mCardinal-loaded NGs (mC_NGs): as a nanoparticle control to account for delivery vehicle effects.
4. PSMA_NT-PGC1α_NG: the targeted treatment.

Mice were injected (subcutaneously) three times per week with NGs (mRNA dose: 0.25mg/kg) for up to 32 days. As shown in Figure 6b,c,e, treatment with NT-PGC1α_NGs, irrespective of their targeting ability, reduced tumor growth significantly in comparison to the untreated or the mCardinal-control treated xenografts. By day 32, when the untreated and mCardinal-control treated tumors had reached volumes of 641 mm^3 and 667 mm^3, respectively, those in the targeted NGs group measured only 171 mm^3, showing nearly a fourfold reduction. To further confirm that NT-PGC1α was indeed being expressed in the tumors, the biodistribution of this protein was determined by measuring expression levels across key organs including the heart, lungs, liver, spleen, kidneys, prostate and tumor tissue, by western blotting and RT-qPCR analysis (Figure S17a-h). NT-PGC1α transcripts were primarily detected in the liver, spleen, kidneys, tumors and in the case of the targeted NG-treated animals, in the prostate (Figure S17i). This was consistent with PSMA-mediated delivery of the RNA payload, as observed previously. We note the known limitations of the subcutaneous xenograft model for evaluating prostate-directed delivery: both targeted and untargeted NT-PGC1α_NGs reduced tumour growth in the flank, and the ectopic location cannot fully discriminate PSMA-mediated intraprostatic delivery from EPR-driven flank-tumour accumulation. The anticipated benefit of PSMA targeting - namely prostate-localised and metastatic-lesion delivery, where EPR is insufficient and receptor-mediated uptake becomes dominant - will be evaluated in orthotopic and metastatic models in follow-up work. The present study was designed as a proof-of-efficacy experiment for an mRNA-delivered NT-PGC1α payload rather than as a full preclinical efficacy package; we have therefore flagged the absence of a formal dose-response, survival analysis, and a benchmark-comparator arm as limitations and planned next steps, while emphasising that the targeted-nanogel data here support receptor-competent delivery rather than differential efficacy in this particular model.

To further assess the biological impact of NT-PGC1α expression in the tumor tissues, histological analysis was performed on tumor sections stained for Ki-67, a proliferation marker and TOMM20, a mitochondrial outer membrane protein indicative of mitochondrial mass (Figure 6f,g). Tumors from NT-PGC1α-treated mice exhibited a striking 85 % reduction in Ki-67-positive nuclei, reflecting a strong anti-proliferative effect (Figure 6f,h). In parallel, TOMM20 staining intensity increased by approximately 300 %, indicating a substantial improvement in mitochondrial content, likely driven by NT-PGC1α activity (Figure 6g,i).

Based on prominent transcriptional changes observed in the RNA-seq dataset we determined the effect of PG1α expression on CAMKK2 protein level. As shown in Figure S18, western blot analysis showed a significant decrease in CAMKK2 protein expression following overexpression of NT-PGC1α. Previous studies reported that CAMKK2 is up-regulated in human PCa and that it acts as a tumor promoter.[11,76–78] Given the role of CAMKK2 as an AR-cooperating, lipogenic tumour-promoter that integrates androgen signalling with energy metabolism, its repression by NT-PGC1α provides a direct mechanistic link between metabolic rewiring and growth suppression in PCa cells. Taken together, the coordinated suppression of cell-cycle drivers, oncogenic regulators (MYC, KRAS, AURKA, CTNNB1) and the AR axis (AR, ARLNC1, NKX3-1) alongside upregulation of CDKN1A/CDKN1B places these transcriptional changes within an established AR–metabolism axis that is mechanistically coherent with NT-PGC1α-driven mitochondrial rewiring, as previously characterised by our group and others[78]. The xenograft Ki-67 reduction (Figure 6f,h; ~85%) provides the tumour-level functional readout of the cell-cycle signature, and the CAMKK2 western blot (Figure S18a,b) provides protein-level confirmation of the AR-metabolism node.[78] In addition, cleaved caspase-3 levels in tumour lysates were comparable across all treatment groups (Figure S18a,c), indicating that the NT-PGC1α-driven tumour-volume reduction proceeds without a measurable increase in apoptotic cell death and is consistent with a predominantly cytostatic mechanism. The mechanistic backbone of NT-PGC1α-driven, AMPK/CAMKK2-linked metabolic rewiring in prostate cancer has been established in our group's prior work[11,78] and the present transcriptomic and protein-level data are consistent with that framework.

Finally, to ascertain whether the animals demonstrated any NG-related systemic toxicity, the NGs were incubated with red blood cells and did not trigger any apparent haemolysis (Figure S19a). Furthermore, serum levels of hepatic enzymes alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were measured following treatment (Figure S19b).[79] While ALT levels remained comparably low across all groups, a modest elevation in AST was observed in the cohort treated with the PSMA_NT-PGC1α_NGs. Although this doesn't reach the levels associated with acute hepatotoxicity (1000 U/L), it could reflect mild hepatic stress in response to NT-PGC1α overexpression. Given the role of PGC1α in mitochondrial biogenesis and oxidative metabolism, it is plausible that elevated systemic expression could transiently impact liver mitochondrial dynamics, leading to slight perturbations in transaminase activity. No overt signs of liver damage or behavioural toxicity were observed in treated animals over the 32-day dosing period. We acknowledge, however, that the present toxicology package - haemolysis, serum ALT/AST and longitudinal body-weight monitoring is necessarily preliminary. A full toxicology workup, including histopathology of liver, kidney, spleen and prostate, complete blood counts and broader serum-chemistry panels, cytokine profiling for innate immune activation, and a longer follow-up window to capture delayed effects, will be required before any clinical-translation claim can be made and is planned for a dedicated follow-up safety study.

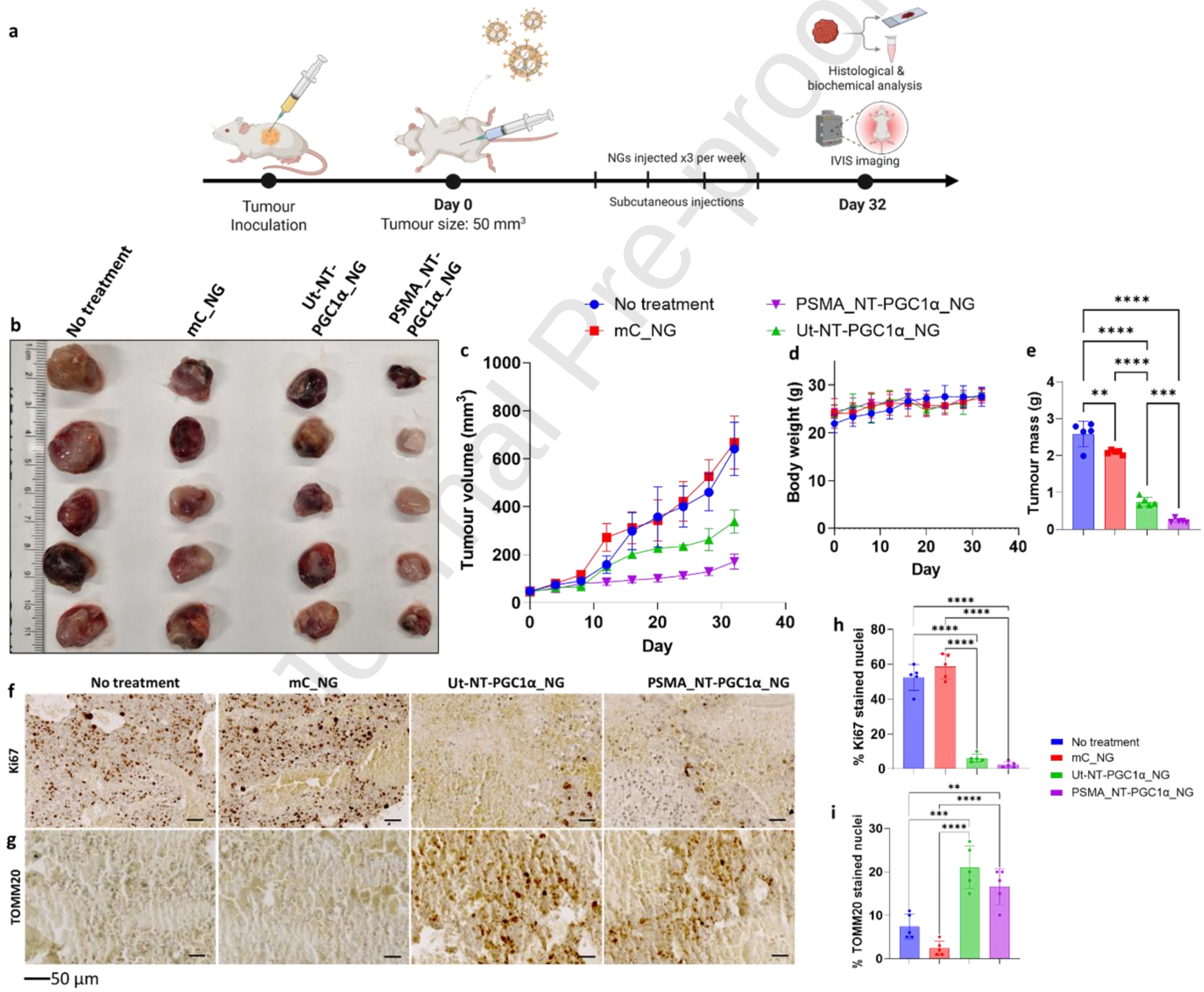

Figure 6: NT-PGC1α overexpression shows enhanced therapeutic efficacy in subcutaneous PCa tumor. a) Schematic presentation of timeline for PCa xenograft generation, NG treatment regime and downstream analysis using harvested animal tissue. b) Images of dissected tumors from conditions including untreated, mC_NG, Ut_NT-PGC1α_NG and PSMA_NT-PGC1α_NG. NGs were injected thrice weekly subcutaneously (n = 5). c) Tumor growth curves and d) changes in body weight of the treated mice. e) Bar graph comparing mass of tumors dissected on day 32 for different conditions. f,g)

Representative images of Ki-67 and TOMM20-stained sections taken from xenografts isolated from mice with various treatments highlighted above are shown, respectively. Scale bar at 50 µm. h,i) Quantification of the percentage of Ki-67- and TOMM20-positive cells in stained sections. Data is shown as mean ± SD from n = 5, and statistical significance was determined using a one-way ANOVA with Dunnett's multiple comparisons test: ****$p < 0.0001$, ***$p < 0.0002$, **$p < 0.0011$, *$p < 0.0322$, ns < not significant.

CONCLUSION

In this study we describe the development and validation of redox-responsive PSMA-targeted nanogels (NGs) for the selective delivery of potential mRNA therapeutics to prostate cancer (PCa) cells, establishing a strong foundation for future clinical translation. The targeted NGs were rationally designed and thoroughly characterized, exhibiting largely homogeneous physicochemical properties and reproducible *in vitro* activity. Uptake dynamics of the targeted NGs were rigorously examined using 3D PCa spheroids, providing insight into performance within complex tumor architecture. These studies could be complemented in the future by advanced imaging approaches, such as light-sheet or confocal microscopy with radial intensity profiling ($d_{1/2}$) to quantify penetration efficiency and intratumoral retention in an even more controlled and reproducible manner.[49]

The NGs achieved tumour-preferential accumulation following clinically relevant systemic administration, with a statistically significant increase in prostate- and tumour-localised mRNA and protein expression relative to untargeted controls (Figure 4 and Figure S13), against a background of appreciable but expected reticuloendothelial and renal exposure that is consistent with all systemically administered nanocarriers. Opportunities for further optimisation include investigating the influence of size, charge and peptide surface coverage, or coupling an anti-PSMA nanobody to enhance receptor-mediated uptake. Systemic administration was chosen over intratumoral injection to reflect real-world therapeutic practice, particularly for regimens requiring repeated dosing.

A key finding of this work is the potent functional activity of PGC1α and its N-terminal isoform NT-PGC1α in prostate cancer models. To date, the tumor-suppressive potential of PGC1α in PCa has remained underexplored, and no studies have demonstrated effects of the magnitude observed here. These results highlight an emerging biological axis and further mechanistic investigations will help to fully elucidate the molecular pathways through which NT-PGC1α exerts its therapeutic effects.

In vivo, systemic delivery of NT-PGC1α-loaded NGs in a xenograft model produced a measurable and well-tolerated reduction in tumor burden. Building on this promising result, future work could explore higher dosing, combination therapies (e.g. with enzalutamide), and long-term evaluation in immunocompetent or humanized mouse models to clarify immunological responses, metabolic fate and durability of tumor control.[78] In parallel, studies employing more physiologically relevant prostate cancer models, such as orthotopic tumors, metastatic models, or patient-derived xenografts, would provide deeper insight into intraprostatic delivery barriers, tumor–microenvironment interactions, and the *in vivo* relevance of PGC1α-mediated tumor suppression. Collectively, these findings underscore a highly innovative and adaptable PSMA-targeted NG platform with significant potential for refinement and clinical translation as a next-generation therapeutic for prostate cancer. In the context of the broader prostate-cancer therapeutic landscape, the present platform is complementary to - rather than directly substitutive for - PSMA-targeted small-molecule radioconjugates such as ^{177}Lu-PSMA-617[35], PSMA antibody-drug conjugates, and lipid-nanoparticle mRNA platforms in clinical or late-preclinical development: it combines receptor-mediated tumour-preferential delivery with redox-responsive intracellular release of a protein-coding mRNA payload, enabling repeat-dose metabolic reprogramming rather than single-modality cytotoxic or radiolytic killing. Advancing the platform to first-in-human studies will require, in addition to the orthotopic, dose-response and survival studies noted above, a dedicated GLP-compliant safety package (histopathology, haematology, cytokine and broader serum-chemistry panels), CMC-grade reproducibility and scale-up of the in situ nanopolymerisation, and regulatory engagement on the disulphide-crosslinked carrier and PSMA-peptide conjugation chemistry.

MATERIALS AND METHODS

Nuclear magnetic resonance. ^{1}H and ^{13}C NMR spectra were recorded using a Bruker AV 400 MHz spectrometer at room temperature. ^{1}H NMR spectra were recorded and chemical shifts are expressed in parts per million (ppm) relative to tetramethylsilane (TMS), using the residual solvent signal as an internal reference.

Fourier-Transform infrared spectra. Fourier transform infrared spectra (FT-IR) was collected using Agilent technologies Cary 630 FTIR. Measurements were performed at room temperature in the wavenumber range of 650cm^{-1} to 4000cm^{-1}. Air was used as the background reference.

Mass spectrometry. Matrix-assisted laser desorption/ionization time-of-flight (MALDI-ToF) mass spec trometry data was collected using Waters® Micromass® MALDI microMX™ MALDI Q-ToF instrument.

Dynamic light scattering (DLS) and electrophoretic light scattering (ELS). Dynamic light scattering (DLS) was used to measure NG size, polydispersity index, whereas zeta-potential was determined via electrophoretic light scattering (ELS). For DLS, the Malvern Zetasizer Ultra (Imperial College London) instrument with backscatter detection at 173° (n = 3) was used. Each run measured 25 subscanning cycles. Before loading the sample, it was filtered using MF-Millipore™ membrane filter (0.45μm pore size, Merck Millipore, UK). The size measurements were taken at 22◦C for each NG synthesized. ζ-potential measurements were carried out using a Zetasizer Nano-ZS instrument with a folded capillary cell in RNase-free water (pH 7.4). A total of 50 subscans were acquired per sample at 22 °C, and the applied voltage was set to 100V, following recommended conditions to minimise Joule heating and improve stability.

Transmission electron microscopy (TEM). NG suspension in water (4μL: 1mgmL−1) was pipetted directly onto a glow discharged lacey carbon film grid with 300 mesh copper (Agar Scientific, UK) and left on the grid for 1min. Then the remainder of the solution was removed gently using a filter paper. The grid was then stained with 4μL of 2% (w/w) uranyl acetate for 1min, after which the stain was removed gently using a filter paper. Samples were imaged on a Talos F200X G2 Transmission Electron Microscope at 45,000 x.

UV-vis absorption spectroscopy and UV-vis fluorescence spectrophotometry. UV–vis absorption spectra of the NGs (0.1mgmL^{-1}) were recorded using an Agilent, Cary 60 UV–vis spectrophotometer over the wavelength range of 200nm to 800nm. All measurements were conducted at room temperature. NG samples were dispersed in deionised water and measured in a quartz cuvette with a 1cm path length. Instrument parameters such as excitation/emission slit widths, scan speed, and PMT voltage were optimized as required. Baseline correction was performed using the corresponding solvent or buffer as a blank.

SPARTA Raman Micro-Spectroscopy Analysis. Raman spectra were acquired using the SPARTA (Single Particle Automated Raman Trapping Analysis) platform on an alpha300R+ confocal Raman micro-spectroscope (WITec, Ulm, Germany), equipped with a 785nm Toptica XTRA II laser (SPARTA Biodiscovery). A 63×/1.0 NA water immersion objective lens (W Plan-Apochromat, Zeiss, Oberkochen, Germany) was employed for sample interrogation. Raman scattered light was directed through a 100 µm optical fibre to a UHTS 300 spectrograph (600grooves/mm, WITec) and detected using a thermoelectrically cooled, back-illuminated CCD camera (iDus DU401-DD, Andor, Belfast, UK), offering a spectral resolution of 3cm^{-1}. The laser power at the sample was maintained at 85mW, and remote control of the laser was performed via serial interface using custom MATLAB scripts (version 2016b, MathWorks, MA, USA). For sample preparation, 200µL of NG solution was used. Ideal particle concentrations ranged from 1×10^{10} to 1×10^{12} particles/mL. Samples were mounted by pipetting onto a 22mm coverslip placed on a standard microscope slide and secured with a droplet of PBS. Measurements were conducted under the water immersion objective. Spectral preprocessing included truncation of the raw spectra to 350–1825 cm^{-1} to eliminate the excitation peak, with the spectral centre set to 1000 cm^{-1}. Cosmic ray spikes were removed using an automated algorithm based on peak intensity and second derivative analysis, followed by manual inspection. Background correction was performed via Whittaker smoothing (λ =100,000), and spectral noise was reduced using a 3-point, first-order Savitzky–Golay filter. Spectra were normalized to the total area under the curve, except for sizing analyses, which relied on perchlorate peak ratio normalization. Subsequent multivariate analyses, including hierarchical cluster analysis (HCA), principal component analysis (PCA), multivariate curve resolution (MCR), and partial least squares discriminant analysis (PLS-DA), were carried out using the software on the instrument.

PSMA-peptide synthesis and characterization. PSMA-617–derived peptide (50 mg, 0.06 mmol) (synthesized by Pepmic, China) was dissolved in anhydrous DCM (5 mL) under a nitrogen atmosphere. To this solution, N-hydroxysuccinimide (NHS)-activated PEG linker bearing a free thiol group (182 mg, 0.18 mmol) dissolved in anhydrous DMF (1 mL) was added, followed by triethylamine (NEt_3, 3 equiv.) (Figure S1). The reaction mixture was stirred at room temperature for 3 h and reaction progression was monitored by TLC. Global deprotection of tert-butyl–based protecting groups was performed by treating the purified intermediate with a mixture of TFA/H_2O (95:5, v/v) for 2 h at room temperature. The solvent was removed under a gentle stream of nitrogen, and residual TFA was co-evaporated with toluene. Lyophilization of the product-containing fractions afforded the final thiolated PEG-modified PSMA targeting construct (PSMA-PEG-SH) as a pale yellow oil. Product identity and purity were confirmed by NMR, HPLC and mass spectrometry.

mRNA-loaded NG synthesis. RNA-loaded nanogels were synthesized with slight modifications to a previously reported protocol, aimed at minimizing RNA degradation.[22,24] Prior to the procedure, all equipment was UV-sterilized and oven-dried overnight to ensure RNAse-free conditions. Surfaces were thoroughly cleaned with RNAseZap (ThermoFisher), and RNAse-free, nitrogen-degassed water was used throughout the synthesis. Briefly, 6 µg of RNA was added to a 1mL reaction solution containing acrylamide (AM; 2 mg, 0.03 mmol), a disulfide-based crosslinker (4.5 mg, 0.01 mmol), 2-[(acryloyloxy)ethyl]trimethylammonium chloride (AETC) or an alternative cationic monomer (2 µL, 0.01 mmol) and mPEG(2kDa)–acrylate (100 µg, 0.05 µmol). For targeted NG, acrylate–PEG(1kDa)–maleimide (0.1% mol) was added to the monomer mixture. The mixture was stirred on ice for 30 minutes under a nitrogen atmosphere. For fluorescently labelled NGs, fluorescein o-acrylate or nile blue acrylamide (0.2 mg, 0.01 mmol) was added, and the reaction was protected from light. Micelle formation was induced by the addition of sodium dodecyl sulfate (SDS; 8 mg, 0.03 mmol), followed by a further 30-minute stirring period. Polymerization was initiated by adding ammonium persulfate (APS; 2.28 mg, 0.01 mmol) and N,N,N′,N′ tetramethylethylenediamine (TEMED; 2.5 µL, 0.02 mmol) in 1mL of RNAse-free, deoxygenated water. The reaction mixture developed a pale blue opalescence within 30 minutes, indicating successful polymerization. Polymerization was terminated by exposure to air once opalescence appeared. The resulting NGs were purified via centrifugation at 1200 xg for 10 minutes using 15 mL Amicon® Ultra-15 centrifugal filter units with a 300 kDa MWCO (Sigma Aldrich, UK). The samples were washed five times with RNAse-free water and stored at 2-8 ºC until further use. The amount of RNA was quantified using the Quant-iT® RNA assay kit, according to the manufacturer's instructions.

PSMA-targeted NG synthesis. Lyophilized PSMA-peptide (PSMA-617-derivative was synthesized by PepMic, China) was reconstituted in DMSO and various concentrations were added to maleimide-coated nanogels (10mgmL^{-1}) in aqueous buffer (pH 6.5–7.5). The conjugation reaction was performed overnight at room temperature under gentle rotation to allow for efficient thiol–maleimide coupling. Following incubation, the reaction mixture was purified by centrifugal filtration using 2 mL Amicon® Ultra-2 centrifugal filter units (50 kDa MWCO, Merck Millipore, UK). Samples were washed three times with Milli-Q water at 4000 xg, 4 ºC for 10 minutes per wash to remove unreacted peptide and byproducts. To assess the extent of conjugation, the residual maleimide content on the nanogels was quantified post-coupling using a colorimetric maleimide quantification assay (MAK162, Sigma-Aldrich, UK). Based on this data, as well as the nanogel concentration (approximated as 1mgmL^{-1}) and particle size, the surface density of maleimide groups was calculated using Equations 1 and 2.[37]

Equation 1: $\#mal/_{NG} = \frac{4}{3}\pi r^3.[mal].\rho.N_A$

Equation 2: $\#mal/_{nm^2} = \frac{\#mal/NG}{4\pi r^2}$

where r is the nanogel radius (in metres), [mal] is the maleimide concentration (mol/m^3), ρ is the nanogel density (assumed as 1.0 g/cm^3 or 1.0 × 10^9 mg/m^3), and NA is Avogadro's number (6.022 x 10^{23} mol^{-1}). To estimate the number and surface density of PSMA-peptides on the nanogels, absorbance measurements at 280mm were taken before and after peptide conjugation. The differential absorbance (ΔA280), attributable to the naphthylalanine residue, was used to determine peptide concentration via the Beer–Lambert law:

Equation 3: $[\text{PSMA peptide}] = \frac{\Delta A_{280}}{l.\varepsilon_{280}}$

where $\varepsilon 280 = 4395.44\ M^{-1}cm^{-1}$ is the molar extinction coefficient of the peptide and $l = 0.1$ cm is the path length. Using the peptide concentration [PSMA peptide] in place of [mal], the number of PSMA peptides per nanogel and their surface density were calculated using analogous equations

Equation 4: $$^{\#pep}/_{NG} = \frac{4}{3}\pi r^3.[PSMA\ peptide].\rho.N_A$$

Equation 5: $$^{\#pep}/_{nm^2} = \frac{\#pep/NG}{4\pi r^2}$$

***In vitro* transcription of RNA.** mRNA encoding all transcripts used in the work was transcribed *in vitro* from PCR-amplified DNA templates. Template for NT–PGC1α and mCardinal were generated by PCR from the following plasmids: mCardinal (Addgene 51311) and PGC1α (Addgene 10974). The DNA templates for NT PGC1α were produced by polymerase chain reaction of the respective plasmids and adding T7 promoter sequence through PCR extension wherever necessary (BioMixTM Red, Bioline). After each PCR step, purification and gel extraction were performed (QIAquick Gel Extraction Kit, Qiagen), according to the manufacturer's instructions. T7 promoter sequences were introduced during PCR amplification using BioMix™ Red polymerase (Bioline). Following each PCR reaction, the products were purified and gel-extracted using the QIAquick Gel Extraction Kit (Qiagen), following the manufacturer's instructions. *In vitro* transcription was performed using the mMESSAGE Mmachine™ T7 Transcription Kit (Invitrogen) according to the manufacturer's protocol. The resulting mRNA was subsequently polyadenylated using the Poly(A) Tailing Kit (Invitrogen). Final purification of the mRNA was carried out using the RNeasy Mini Kit (Qiagen). RNA concentration and purity were assessed using a NanoDrop One spectrophotometer (ThermoFisher, UK), and samples were stored at −80 ºC until use. Each plasmid DNA was transformed into E. coli 5α (New England BioLabs, UK) using standard heat-shock treatment. Following a 1 h recovery at 37 ºC with agitation, the bacteria were plated on Luria broth (LB) agar plates containing ampicillin (100 μgmL^{-1}) (Sigma-Aldrich, UK) and incubated overnight at 37 ºC. Single colonies were picked the next day and used to inoculate 10 mL of liquid LB medium containing 100 μgmL^{-1} ampicillin. Cultures were grown at 37 ºC for 5 h, after which 200 µL of the suspension was transferred to 200 mL of fresh LB medium supplemented with ampicillin and cultured to saturation. Plasmid DNA was extracted using the Plasmid Plus MaxiPrep Kit (Qiagen, UK) following the manufacturer's protocol. The concentration and purity of the purified plasmid DNA were assessed using a NanoDrop One spectrophotometer (ThermoFisher, UK), and samples were stored at −20 ºC.

Cell culture. C4-2 and C4-2b cell lines were sourced from the American Type Culture Collection (ATCC). Cells were cultured on flasks and plates coated with poly-L-lysine (0.1 mg/mL; Thermo, UK) in RPMI 1640 medium (Thermo, UK), supplemented with 10% fetal bovine serum (FBS) and 1% penicillin–streptomycin (Sigma-Aldrich, UK). For all experiments, C4-2 cells were used at passage numbers below 25. All cell lines were maintained at 37 ºC in a humidified incubator with 5 % CO_2.

Transfection studies. C4-2 cells were seeded in 96-well plates at a density of 1×10^4 cells/well, in triplicate for each condition. Cells were allowed to adhere overnight prior to the experiment. The plates were placed in the IncuCyte instrument, housed within a standard humidified incubator maintained at 37 ºC and 5 % CO_2. Cells were treated with various concentrations of AETC_NGs suspended in complete culture medium containing 10 % FBS. Where applicable, transfection efficiency was benchmarked against two commercial reagents – Polyplus® jetMESSENGER® and Lipofectamine™ MessengerMax™ – each prepared following the manufacturer's protocol using 3.25 µg of RNA. Automated phase-contrast and green fluorescence imaging was performed using the IncuCyte system. Images were acquired every 1 h over a total period of 86 h, with four images captured per well (image resolution: 1.22 µm/pixel). Green fluorescence was imaged using the default exposure time of 800 ms. Initial image analysis was optimized using untreated control wells to establish a threshold for background fluorescence. Processing definitions, including confluence masks for both fluorescence and phase-contrast images, were adjusted to minimise background signal and exclude cellular debris. Once optimal settings were determined, they were applied uniformly across all wells and cell types. GFP fluorescence was automatically normalized to cell density by the IncuCyte software to account for differences in cell number between wells. To quantify total cellular protein content, cells were lysed in RIPA buffer and centrifuged at 14000 rpm for 10 mins at 4 ºC. The resulting supernatant was used to determine protein concentration via the BCA protein assay (ThermoFisher, UK), following the manufacturer's instructions.

Cellular uptake studies via confocal microscopy. C4-2 cells (5×10^4 cells/well) were seeded into µ-Slide 8-well chambered slides (Ibidi, GmbH) and incubated overnight. The following day, the medium was replaced with fresh culture medium containing 20 µg/mL of nanoparticles (NGs), and the cells were incubated for 2-4 h. After treatment, cells were washed three times with PBS and fixed using 4 % paraformaldehyde for 10 mins on ice. Subsequent washes were followed by nuclear staining with Hoechst 33342 (100 ng/mL in PBS) for 15 mins. Cells were again washed thrice with PBS and imaged using a Leica TCS SP8 DLS inverted confocal microscope (Leica, Germany) with LAS X software (Objective lens 63X NA 1.4). Image analysis was performed using ImageJ (NIH, USA).

Mitochondrial biogenesis assessment using flow cytometry. C4-2 cells (2×10^5/well) were seeded in 24-well plates and incubated overnight. Cells were treated with PGC1α_NGs (40 µg/mL), PGC1α_jetM, and/or BI9774 (10 μM). After 24h, NG-treated cells received a second dose of PGC1α_NGs with/without BI9774 for another 24h. At 48h, cells were stained with MitoTracker® Green (MTG) and TMRM for 30 min. After washing and trypsinization, cells were pelleted and washed twice with PBS, resuspended in FACS buffer (1% BSA in PBS), and analyzed using a FACSymphony™ A3 instrument. Data analysis was conducted using FlowJo software to calculate mean fluorescence intensities (MFI)

Immunofluorescence studies. C4-2 cells used for imaging were seeded in poly-L-lysine-treated Ibidi glass bottomed 8-well chamber slides (Ibidi, GmbH) at a seeding density of 50,000 cells/well and were left to adhere overnight prior to the transfection and imaging experiments. The cells were either left untreated or treated with NGs (FLSSNG or FLNG or PGC1α RNA-loaded NGs (40 μg/mL) and in the case of the experiment relating to PGC1α, the cells were treated with or without BI (10 μM). The cells were then fixed in 4% paraformaldehyde for 10 min on ice and washed thrice with PBS. Cells were permeabilized in 0.1% Triton X-100 in PBS for 3 min and washed thrice with PBS. Blocking was performed in 1% BSA in PBS for 1 h, followed by PBS washes and overnight incubation in primary antibody (NT-PGC1α, 1:100 dilution). After washing, secondary antibody Goat anti-Rabbit IgG (H+L) Alexa Fluor™ 568 (1:500) was added for 1 h. The cells were counterstained with Hoechst 33342 (1 μg/mL) for 10 min, washed, and imaged on a Leica TCS SP8 DLS inverted microscope (Leica, Germany) using LASX software. Image analysis was performed using ImageJ (NIH, USA), and colocalization was quantified using JACoP, reporting Pearson's correlation coefficient.

Transfection efficiency quantification by western blot analysis. β2-knockout HEK293T and C4-2 cells (1×10^6) were seeded into 6 cm collagen-coated plates and incubated for 24 h. After replacing with fresh media, cells were treated with various concentrations of β2 or PGC1α-loaded NGs for specific durations. For protein collection, cells were washed twice with ice-cold PBS and lysed using RIPA buffer. Lysates were centrifuged at 14000 rpm for 10 min at 4 ºC, and supernatants were collected. Protein content was quantified using a BCA assay kit (ThermoFisher, UK). Samples were resolved by NuPAGE™ 4–12% Bis-Tris gels and transferred to PVDF membranes. Primary antibodies were used at 1:1000 dilution: total OXPHOS (Abcam, ab110413), vinculin (Sigma, V9131), NT-PGC1α (Santa Cruz, sc-518025), CAMKK2 (Abcam, ab96531) and mCardinal (STJ140229). Secondary antibodies used were LI-COR IRDye (1:10,000) or HRP-conjugated (1:3000), with detection via Odyssey Infrared Imager or Amersham ImageQuant 800. Quantification was done using Odyssey software or ImageJ, and results were expressed as the signal ratio relative to control antibodies.

Transfection efficiency quantification by RT-qPCR. RNA was extracted from human C4-2 cells using TRIzol reagent (Invitrogen) and purified using a RNeasy column (Qiagen). cDNA synthesis was performed using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems™), followed by qPCRusing PowerUp™ SYBR™Green Master Mix. Gene expression was normalized to GAPDH. RT-qPCR was conducted using a CFX Connect system (Bio-Rad, UK), with technical triplicates for each reaction. C_t values were averaged and analyzed using the calibration curve method. All primer sequences used in this study are detailed below:

Table 2.1: RT-qPCR primer sequences (F labelled for forward, R labelled for Reverse)

Target gene	Direction	Sequence (5'-3')
NT-PGC1α	F	TCACACCAAACCCACAGAGA
	R	CTGGAAGATATGGCACAT
mCardinal	F	ACAACGTCAAGCTCAGAGGG
	R	TTCTTTCCAGTCTGCGGTCC
GAPDH	F	ACTCCACTCACGGCAAATTC
	R	TCTCCATGGTGGTGAAGACA

Endocytosis inhibition studies C4-2 cells (1×10^5) were seeded in 24-well plates and incubated overnight. Cells were washed with cold PBS (2×500 µL) and preincubated with the following inhibitors in serum-free media at 37◦C for 1h:

- Hyperosmolar sucrose (45mM, clathrin-mediated endocytosis)
- EIPA (100µM, macropinocytosis)
- Nystatin (30µM, caveolin pathway)
- Dyngo (30µM, clathrin-independent/dynamin-dependent)
- 7-ketocholesterol (30µM, clathrin-independent/dynamin-independent)
- Cytochalasin D (5µM, phagocytosis/macropinocytosis)
- NaN_3/2-deoxy-D-glucose (affects all pathways)
- A control set was kept at 4 ºC to inhibit all energy-dependent processes

Following inhibitor treatment, cells were incubated with 20 µg/mL NGs and the same inhibitors. After incubation, cells were washed with PBS, detached using 0.25% trypsin-EDTA for 3 min at 37 ºC, and prepared for flow cytometry as described above.

Cytotoxicity assessment. C4-2 cells (1×10^4cells/well) were seeded in 96-well plates and incubated overnight. The following day, cells were treated with jetMESSENGER® (0.25 µL, as per manufacturer's instructions for 96-well format) and increasing concentrations of NGs (0–500 µg/mL). After 24 h, cytotoxicity was measured using the Cell Counting Kit-8 (Sigma-Aldrich, UK) following the manufacturer's protocol.

Mitochondrial respiration analysis using the seahorse XF mito stress test. Mitochondrial function in C4–2 cells was evaluated using the Seahorse XF Cell Mito Stress Test Kit (Agilent Technologies) on a Seahorse XF96 Extracellular Flux Analyzer. Cells were seeded at a density of 20,000 cells per well in Seahorse XF DMEM assay medium (Agilent), supplemented with 10 mM Seahorse XF glucose, 2 mM Seahorse XF L-glutamine, and 5% fetal bovine serum (Gibco, 12676), and incubated overnight at 37 ºC in a CO_2-free incubator. On the day of the assay, the medium was replaced with fresh assay medium, and cells were equilibrated for 1 h at 37 ºC in a non-CO_2 incubator. Mitochondrial stress was induced by sequential injection of the following compounds through the Seahorse cartridge ports to reach final concentrations in each well: oligomycin (1.5 µM, port A), FCCP (0.5 µM, port B), and a combination of rotenone and antimycin A (1.0 µM each, port C). This allowed for the measurement of key respiratory parameters including basal respiration, ATP production, maximal respiration, and non-mitochondrial respiration. Each measurement cycle consisted of 3 min mixing, 0 min wait, and 3 min measurement, with three cycles run per condition. Data were collected and analyzed using Seahorse Wave software (version 2.6.1, Agilent Technologies). After the assay, cells were fixed with 4% paraformaldehyde and stained with 0.5% crystal violet. Following washing and drying, the dye was solubilized using 10% acetic acid, and absorbance at 590nm was measured to confirm consistent seeding. Oxygen consumption rate (OCR) values were normalized to crystal violet absorbance.

Haemolysis assay. To assess the potential membrane-disruptive effects of NGs, haemolysis assays were performed using red blood cells (RBCs) isolated from animal blood. Whole blood (6 mL) was diluted with 12 mL PBS and centrifuged at 3000 rpm for 10 mins. The pellet was washed repeatedly with 10 mL PBS until the supernatant became clear. A 2% RBC suspension was prepared in sterile PBS (pH 7.4), and incubated with various concentrations of AETC_NGs (0–500 µg/mL) at 37 ºC. PBS and 10% SDS were used as negative and positive controls, respectively. At 2, 4, 8, 12, and 24 h, samples were centrifuged at 2500 rpm for 5 mins, and 100 µL of supernatant was transferred to a 96-well plate. The absorbance of released haemoglobin was measured at 570nm. Percentage haemolysis was calculated using:

$$\% \, Haemolysis = \frac{Abs_{of\,sample} - Abs_{of\,negative\,control}}{Abs_{of\,positive\,control} - Abs_{of\,negative\,control}} \times 100\%$$

Percent haemolysis values were calculated from three separate experiments.

Immunohistochemical and immunofluorescence staining of tumor sections. Whole tumors were fixed in 4% paraformaldehyde (PFA) overnight at 4 ºC, embedded in paraffin, and sectioned at a thickness of 5 µm. For immunohistochemical analysis, tissue sections were deparaffinized in 100% xylene, rehydrated through a graded ethanol series (100% to 70%), and subjected to antigen retrieval by boiling in sodium citrate buffer (pH 6.0) for 5 min using a pressure cooker. Endogenous peroxidase activity was quenched by incubating the sections in 0.3% hydrogen peroxide (H_2O_2) in distilled water. After washing in Tris-buffered saline (TBS), non-specific binding was blocked by incubating the sections in 10% normal goat serum in TBS for 1 h at room temperature. Primary antibodies against Ki-67 (Abcam, ab16667; 1:200 dilution) and TOMM20 (Abcam, ab186735; 1:100 dilution) were applied overnight at 4 ºC. The following day, sections were washed in TBS containing 0.1% Tween-20 (TBS-T) and incubated for 1 h at room temperature with a biotinylated goat secondary antibody. After further washing in TBS-T, sections were incubated with an avidin–biotin complex (VECTASTAIN® Elite ABC Kit, Vector Laboratories) for 30 min according to the manufacturer's instructions. Staining was visualized using the DAB Substrate Kit (Vector Laboratories), followed by counterstaining with Gill's hematoxylin (Sigma-Aldrich). Sections were then dehydrated, cleared, and mounted using DPX mountant (Sigma-Aldrich) and imaged using the AxioScan Slidescanner, Zeiss. For quantification, Fiji (ImageJ) software was used. Automated cell detection was performed on manually annotated regions using the colour-deconvolved haematoxylin channel. In parallel, immunofluorescence staining was performed on 5 µm frozen tumor sections to validate mCardinal expression. Sections were incubated overnight at 4 ºC with primary antibody against mCardinal (St John's Laboratory, STJ140229; 1:500 dilution), counterstained with DAPI, and imaged to assess expression, as before.

Spheroid formation, imaging, and nanoparticle uptake. C4-2 cells were seeded into NunclonTM Sphera 96-well U-bottom plates (Thermo Fisher Scientific) at densities ranging from 100 to 500 cells per well in 200 µL of complete RPMI-1640 medium supplemented with 10% fetal bovine serum (FBS), 100 U/mL penicillin-streptomycin, and 24.8 µg/mL rat tail collagen (Sigma-Aldrich). Cultures were maintained at 37 ºC in a humidified 5% CO_2 incubator. Media were replenished every 72 h by removing 100 µL of medium from each well and replacing it with an equal volume of fresh growth medium using a multichannel pipette. Spheroid growth and morphology were assessed using the

Incucyte® Spheroid Analysis Software Module. For live imaging, mature spheroids were embedded in an agarose matrix within 8-well Ibidi chamber slides, counterstained with Hoechst 33342 (1 µg/mL), and visualized using a Leica TCS SP8 DLS confocal microscope. Image analysis was performed using Fiji (ImageJ, NIH). For nanoparticle uptake studies, spheroids were cultured for 7 days prior to treatment with nanogels (NGs) at a final concentration of 40 µg/mL. NG penetration was evaluated by acquiring z-stack images across the spheroid depth using confocal microscopy. To quantify fluorescence signal across the spheroid, the integrated density method was employed: for each z-slice, a region of interest (ROI) encompassing the entire spheroid cross-section was manually drawn in Fiji, and the integrated density was measured as the product of the ROI area and the mean gray value. This approach provided a depth-resolved measure of nanoparticle fluorescence, enabling the assessment of NG distribution throughout the spheroidal mass.

mCardinal_NG biodistribution in healthy mice. All animal experiments were performed in accordance with the UK Animals (Scientific Procedures) Act 1986 and approved by the internal ethics committee and UK Home Office under project license PP0283859 and personal license I77452048. Animals were provided food and water *ad libitum*. To evaluate the biodistribution of mCardinal_NGs, C57BL/6 mice received 200 µL injections of nanoparticles (NGs) via either the tail vein or subcutaneous fat in adult male mice. The formulation contained 0.25 mg/kg mCardinal mRNA and 60 mg/kg polymer. Mice were euthanized at predefined time points following administration. Major organs, including brain, heart, liver, spleen, lungs, and kidneys, were harvested, along with tumor tissues when applicable. Fluorescence biodistribution was assessed using an *In vivo* Imaging System (IVIS excitation: 600 nm; emission: 640 nm) to quantify mean fluorescence intensity (MFI).

C4-2b xenograft studies. Male NOD-scid gamma (NSG)-614 mice (6–8 weeks old) were purchased from Charles River (UK) and acclimatized for one week before experimentation. Mice were subcutaneously inoculated on the right flank with 2.5×10^6 C4-2b cells in 100 µL of serum-free medium mixed with 100 µL of Matrigel Basement Membrane Matrix (High Concentration, Corning). Tumors were allowed to grow to approximately 50 mm^3 before initiating treatment. Mice were randomly assigned to five treatment groups: (1) untreated, (2) Untargeted NT_PGC1α_NG, (3) mCardinal_NG, (4) PSMA_NT PGC1α_NG. In all groups, the mRNA dose was 0.25mg/kg, and the polymer was administered at 60mg/kg. NGs were injected subcutaneously three times per week for 32 days. Tumor volume was monitored twice weekly using digital callipers and calculated using:

$$Volume = \frac{\pi}{6} \cdot length \cdot width \cdot height$$

At the end of the study, blood samples were collected for serum analysis to assess potential systemic toxicity.

Serum biochemistry for *in vivo* cytotoxicity. To evaluate systemic toxicity, serum samples obtained from tumor-bearing mice at study endpoint were analyzed for liver enzyme levels at the MRC MDU Mouse Biochemistry Laboratory. Alanine transaminase (ALT) and aspartate aminotransferase (AST) concentrations were measured by the Core 83 Biochemical Assay Laboratory, Cambridge, UK.

RNA sequencing

Total RNA was extracted from C4-2b cells treated with NT-PGC1α-loaded NGs or Empty NGs and cultured in 6-well plates (n=5), using the QiagenTM RNeasy Mini Kit. RNA was subsequently cleaned using the RNA Clean and Concentrator Kit (Zymo Research, Germany) and quantified with a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). A total of 2 µg RNA per sample was diluted in 20 µL of nuclease-free water and analysed on a Bioanalyzer 2100 system (Agilent Technologies, 84 Santa Clara, CA, USA) to assess RNA integrity. All samples exhibited RNA Integrity Numbers (RIN) greater than 9. Single-end sequencing (100 bp reads) was performed using the Illumina NextSeq 2000 platform (Illumina, San Diego, CA, USA) by Mr Ivan Andrew at the MRC LMS Genomics Facility. RNA-seq reads were processed using cutadapt v4.7[80] to remove Illumina adapters, to quality trim at Q20, and to filter read pairs containing N bases or where either read <31 bps. Processed reads were quantified with Salmon v1.9.0[81] using transcripts from the GRCh38 Ensemblv107 annotations. Salmon's expectation maximisation procedures were set to enable modelling of sequencing and GC biases. Within R, transcript-level counts from Salmon were imported and aggregated to the gene level using tximport v1.34.0.[82] Sample quality assessments were performed in relation to collected RNA integrity (RIN) values, sample concentrations used in library preparation (ng/µl), read GC content, and percentage read mapping. Normalisation, PCA-based and clustering based quality control based on normalised values, and differential expression analyses were performed using DESeq2 v1.46.0.[83] Independent hypothesis weighting was applied to optimise the power of p-value filtering using IHW v1.34.0[84], and log2 fold change shrinkage was performed with ashr v2.2-63[85] to reduce the impact of low expression on fold change estimation. Multiple testing correction was applied by passing thresholds for both false discovery rate and log2 fold change within DESeq2. Significance was assessed using a log2 fold change threshold of 1 (i.e., 2-fold change) and a false discovery rate of $q < 0.01$. Downstream gene ontology (GO) enrichment analysis was performed using clusterProfiler v4.14.4[86], targeting the Biological Process (BP) GO terms. Benjamini–Hochberg correction was applied, with GO terms considered significant at $q < 0.05$. The data discussed in this publication have been deposited in NCBI's Gene Expression Omnibus and are accessible through GEO Series accession number GSE335925 (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE335925).

Author Contributions

Rupali Dabas: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Lida Kabir:** Writing – review & editing, Methodology, Data curation. **Naveenan Navaratnam:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Hothri A. Moka:** Data curation, Formal analysis. **Alessandro Sardini:** Writing – review & editing, Data curation. **Guianfranco Coria Mazzei**: Writing – review & editing, Data curation. **Nazila Kamaly:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Conceptualization. **David Carling:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

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References:

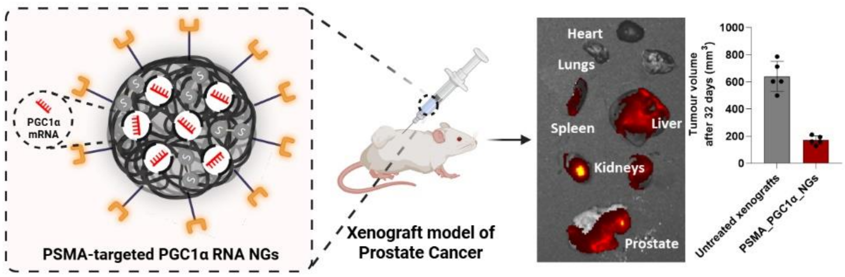
1. Siegel, R. L., Miller, K. D., Fuchs, H. E. & Jemal, A. Cancer statistics, 2022. *CA: A Cancer Journal for Clinicians* **72**, 7–33 (2022).
2. Datta, D., Aftabuddin, M., Gupta, D. K., Raha, S. & Sen, P. Human Prostate Cancer Hallmarks Map. *Scientific Reports* **6**, 30691 (2016).
3. Tang, D. G. Understanding and targeting prostate cancer cell heterogeneity and plasticity. *Seminars in Cancer Biology* **82**, 68–93 (2022).
4. Pavlova, N. N. & Thompson, C. B. THE EMERGING HALLMARKS OF CANCER METABOLISM. *Cell metabolism* **23**, 27–47 (2016).

5. Ahmad, F., Cherukuri, M. K. & Choyke, P. L. Metabolic reprogramming in prostate cancer. *British Journal of Cancer* **125**, 1185–1196 (2021).
6. LeBleu, V. S. *et al.* PGC-1α mediates mitochondrial biogenesis and oxidative phosphorylation to promote metastasis. *Nature cell biology* **16**, 992–15 (2014).
7. Torrano, V. *et al.* The metabolic co-regulator PGC1α suppresses prostate cancer metastasis. *Nat Cell Biol* **18**, 645–656 (2016).
8. Valcarcel-Jimenez, L. *et al.* PGC1α Suppresses Prostate Cancer Cell Invasion through ERRα Transcriptional Control. *Cancer Research* **79**, 6153–6165 (2019).
9. Ahmad, F., Cherukuri, M. K. & Choyke, P. L. Metabolic reprogramming in prostate cancer. *Br J Cancer* **125**, 1185–1196 (2021).
10. Cutruzzolà, F. *et al.* Glucose Metabolism in the Progression of Prostate Cancer. *Front. Physiol.* **8**, (2017).
11. Penfold, L. *et al.* AMPK activation protects against prostate cancer by inducing a catabolic cellular state. *Cell Rep* **42**, 112396 (2023).
12. Kulkarni, J. A. *et al.* The current landscape of nucleic acid therapeutics. *Nature Nanotechnology* **16**, 630–643 (2021).
13. Qin, S. *et al.* mRNA-based therapeutics: powerful and versatile tools to combat diseases. *Signal Transduction and Targeted Therapy* **7**, 1–35 (2022).
14. Rohner, E., Yang, R., Foo, K. S., Goedel, A. & Chien, K. R. Unlocking the promise of mRNA therapeutics. *Nature Biotechnology* **40**, 1586–1600 (2022).
15. Hou, X., Zaks, T., Langer, R. & Dong, Y. Lipid nanoparticles for mRNA delivery. *Nature Reviews Materials* **6**, 1078–1094 (2021).
16. Ju, Y. *et al.* Impact of anti-PEG antibodies induced by SARS-CoV-2 mRNA vaccines. *Nature Reviews Immunology* **23**, 135–136 (2023).
17. Morgulchik, N. & Kamaly, N. Meta-analysis of In Vitro Drug-Release Parameters Reveals Predictable and Robust Kinetics for Redox-Responsive Drug-Conjugated Therapeutic Nanogels. *ACS Applied Nano Materials* **4**, 4256–4268 (2021).
18. Ashwani, P. V. *et al.* Stimuli-Responsive and Multifunctional Nanogels in Drug Delivery. *Chemistry & Biodiversity* **20**, e202301009 (2023).
19. Chakroborty, S. *et al.* Stimuli-responsive nanogels: A smart material for biomedical applications. *Journal of Molecular Liquids* **403**, 124828 (2024).
20. Basak, S. *et al.* Simultaneous Cross-Linking and Cross-Polymerization of Enzyme Responsive Polyethylene Glycol Nanogels in Confined Aqueous Droplets for Reduction of Low-Density Lipoprotein Oxidation. *Biomacromolecules* **22**, 386–398 (2021).
21. Blagojevic, L. & Kamaly, N. Nanogels: A chemically versatile drug delivery platform. *Nano Today* **61**, 102645 (2025).
22. Dabas, R. *et al.* Precise intracellular uptake and endosomal release of diverse functional mRNA payloads via glutathione-responsive nanogels. *Materials Today Bio* **30**, 101425 (2025).
23. Dabas, R., Koh, A., Carling, D., Kamaly, N. & Brown, A. E. X. Redox-Responsive Polymeric Nanogels as Efficient mRNA Delivery Vehicles in Caenorhabditis elegans. *microPublication Biology* https://doi.org/10.17912/micropub.biology.001428 (2024) doi:10.17912/micropub.biology.001428.
24. Dabas, R., Blagojevic, L. & Kamaly, N. Synthesis of Stimuli-responsive Nanogels using Aqueous One-step Crosslinking and Co-nanopolymerization. *Journal of Visualized Experiments (JoVE)* e63981 (2025) doi:10.3791/63981.
25. Dabas, R. & Kamaly, N. Redox-Responsive Nanogels for Precision Protein Delivery. *European Polymer Journal* **215**, 113183 (2024).
26. Huang, K. *et al.* Small, Traceable, Endosome-Disrupting, and Bioresponsive Click Nanogels Fabricated via Microfluidics for CD44-Targeted Cytoplasmic Delivery of Therapeutic Proteins. *ACS Applied Materials & Interfaces* **11**, 22171–22180 (2019).

27. Zhao, X. *et al.* Redox-Sensitive Gelatin/Silica-Aptamer Nanogels for Targeted siRNA Delivery. *Nanoscale Research Letters* **14**, 273 (2019).
28. Huynh, U. *et al.* Targeted Drug Delivery Using a Plug-to-Direct Antibody–Nanogel Conjugate. *Biomacromolecules* https://doi.org/10.1021/acs.biomac.2c01269 (2023) doi:10.1021/acs.biomac.2c01269.
29. Debnath, S. *et al.* PSMA-Targeting Imaging and Theranostic Agents—Current Status and Future Perspective. *International Journal of Molecular Sciences* **23**, 1158 (2022).
30. Wang, F., Li, Z., Feng, X., Yang, D. & Lin, M. Advances in PSMA-targeted therapy for prostate cancer. *Prostate Cancer Prostatic Dis* **25**, 11–26 (2022).
31. Sartor, O. *et al.* Lutetium-177–PSMA-617 for Metastatic Castration-Resistant Prostate Cancer. *New England Journal of Medicine* **385**, 1091–1103 (2021).
32. Ajish Kumar, K. S. & Mathur, A. Total chemical synthesis of PSMA-11: API for 68Ga-PSMA-11 used for prostate cancer diagnosis. *European Journal of Medicinal Chemistry Reports* **3**, 100014 (2021).
33. Schäfer, M. *et al.* Structure–Activity Relationships and Biological Insights into PSMA-617 and Its Derivatives with Modified Lipophilic Linker Regions. *ACS Omega* **10**, 7077–7090 (2025).
34. Chang, S. S. Overview of Prostate-Specific Membrane Antigen. *Reviews in Urology* **6**, S13–S18 (2004).
35. Kumar, K. S. A. & Mathur, A. A convenient total synthesis of PSMA-617: A prostate specific membrane antigen (PSMA) ligand for prostate cancer endotherapeutic applications. *European Journal of Medicinal Chemistry Reports* **6**, 100084 (2022).
36. Lundmark, F. *et al.* Design, Synthesis, and Evaluation of Linker-Optimised PSMA-Targeting Radioligands. *Pharmaceutics* **14**, 1098 (2022).
37. Bazban-Shotorbani, S. *et al.* Effect of Nanoparticle Biophysicochemical Properties on Binding and Transport across Cardiovascular Endothelial Dysfunction Models. *ACS Appl. Nano Mater.* **4**, 4077–4091 (2021).
38. Penders, J. *et al.* Single Particle Automated Raman Trapping Analysis. *Nature Communications* **9**, 4256 (2018).
39. Ginalski, K., Grishin, N. V., Godzik, A. & Rychlewski, L. Practical lessons from protein structure prediction. *Nucleic Acids Research* **33**, 1874–1891 (2005).
40. Jr, G. J. T. RAMAN SPECTROSCOPY OF PROTEIN AND NUCLEIC ACID ASSEMBLIES. *Annual Review of Biophysics* **28**, 1–27 (1999).
41. Bakht, M. K. & Beltran, H. Biological determinants of PSMA expression, regulation and heterogeneity in prostate cancer. *Nature Reviews Urology* **22**, 26–45 (2025).
42. Sheehan, B. *et al.* Prostate-Specific Membrane Antigen Expression and Response to DNA Damaging Agents in Prostate Cancer. *Clinical Cancer Research* **28**, 3104–3115 (2022).
43. Rennick, J. J., Johnston, A. P. R. & Parton, R. G. Key principles and methods for studying the endocytosis of biological and nanoparticle therapeutics. *Nature Nanotechnology* **16**, 266–276 (2021).
44. Rajasekaran, S. A. *et al.* A Novel Cytoplasmic Tail MXXXL Motif Mediates the Internalization of Prostate-specific Membrane Antigen. *Molecular Biology of the Cell* **14**, 4835–4845 (2003).
45. Cultrara, C. N. *et al.* Size Matters: Arginine-Derived Peptides Targeting the PSMA Receptor Can Efficiently Complex but Not Transfect siRNA. *Molecular Therapy. Nucleic Acids* **18**, 863–870 (2019).
46. Rivero-Buceta, E. *et al.* PSMA-Targeted Mesoporous Silica Nanoparticles for Selective Intracellular Delivery of Docetaxel in Prostate Cancer Cells. *ACS Omega* **4**, 1281–1291 (2019).
47. Anilkumar, G. *et al.* Prostate-specific Membrane Antigen Association with Filamin A Modulates Its Internalization and NAALADase Activity1. *Cancer Research* **63**, 2645–2648 (2003).
48. Anilkumar, G. *et al.* Association of prostate-specific membrane antigen with caveolin-1 and its caveolae-dependent internalization in microvascular endothelial cells: Implications for targeting to tumor vasculature. *Microvascular Research* **72**, 54–61 (2006).
49. Niora, M. *et al.* Head-to-Head Comparison of the Penetration Efficiency of Lipid-Based Nanoparticles into Tumor Spheroids. *ACS Omega* **5**, 21162–21171 (2020).

50. Chu, J. *et al.* Non-invasive intravital imaging of cellular differentiation with a bright red-excitable fluorescent protein. *Nature methods* **11**, 572–578 (2014).
51. Zhang, W. *et al.* Theranostic Bottle-Brush Polymers Tailored for Universal Solid-Tumor Targeting. *ACS Nano* **18**, 11688–11702 (2024).
52. He, Z. *et al.* A Multidimensional Approach to Modulating Ionizable Lipids for High-Performing and Organ-Selective mRNA Delivery. *Angewandte Chemie International Edition* **n/a**, e202310401.
53. Kumar, M., Kulkarni, P., Liu, S., Chemuturi, N. & Shah, D. K. Nanoparticle biodistribution coefficients: A quantitative approach for understanding the tissue distribution of nanoparticles. *Advanced Drug Delivery Reviews* **194**, 114708 (2023).
54. Tenchov, R., Bird, R., Curtze, A. E. & Zhou, Q. Lipid Nanoparticles–From Liposomes to mRNA Vaccine Delivery, a Landscape of Research Diversity and Advancement. *ACS Nano* **15**, 16982–17015 (2021).
55. Golombek, S. K. *et al.* Tumor targeting via EPR: Strategies to enhance patient responses. *Advanced Drug Delivery Reviews* **130**, 17–38 (2018).
56. Zhang, Y. *et al.* Alternative mRNA Splicing Produces a Novel Biologically Active Short Isoform of PGC-1α. *The Journal of Biological Chemistry* **284**, 32813–32826 (2009).
57. Malik, N. *et al.* Induction of lysosomal and mitochondrial biogenesis by AMPK phosphorylation of FNIP1. *Science* **380**, eabj5559 (2023).
58. Kholmukhamedov, A., Schwartz, J. M. & Lemasters, J. J. MitoTracker Probes and Mitochondrial Membrane Potential. *Shock (Augusta, Ga.)* **39**, 543 (2013).
59. Torrano, V. *et al.* The metabolic co-regulator PGC1α suppresses prostate cancer metastasis. *Nat Cell Biol* **18**, 645–656 (2016).
60. Zhang, Z. *et al.* Key genes involved in cell cycle arrest and DNA damage repair identified in anaplastic thyroid carcinoma using integrated bioinformatics analysis. *Transl Cancer Res* **9**, 4188–4203 (2020).
61. Xu, Y., Wu, X., Li, F., Huang, D. & Zhu, W. CDCA4, a downstream gene of the Nrf2 signaling pathway, regulates cell proliferation and apoptosis in the MCF-7/ADM human breast cancer cell line. *Mol Med Rep* **17**, 1507–1512 (2018).
62. Shen, T. & Huang, S. The Role of Cdc25A in the Regulation of Cell Proliferation and Apoptosis. *Anti-Cancer Agents in Medicinal Chemistry* **12**, 631–639 (2012).
63. Zhang, C. *et al.* Down-regulation of CCNE1 expression suppresses cell proliferation and sensitizes gastric carcinoma cells to Cisplatin. *Biosci Rep* **39**, BSR20190381 (2019).
64. Pawlonka, J., Rak, B. & Ambroziak, U. The regulation of cyclin D promoters - review. *Cancer Treat Res Commun* **27**, 100338 (2021).
65. Leal-Esteban, L. C. & Fajas, L. Cell cycle regulators in cancer cell metabolism. *Biochim Biophys Acta Mol Basis Dis* **1866**, 165715 (2020).
66. Viner-Breuer, R., Yilmaz, A., Benvenisty, N. & Goldberg, M. The essentiality landscape of cell cycle related genes in human pluripotent and cancer cells. *Cell Div* **14**, 15 (2019).
67. Liu, Z., Sun, Q. & Wang, X. PLK1, A Potential Target for Cancer Therapy. *Transl Oncol* **10**, 22–32 (2017).
68. Qin, L.-T. *et al.* Clinical value and potential mechanisms of BUB1B up-regulation in nasopharyngeal carcinoma. *BMC Med Genomics* **15**, 272 (2022).
69. Arroyo, M. *et al.* TOP2A inhibition and its cellular effects related to cell cycle checkpoint adaptation pathway. *Sci Rep* **15**, 3831 (2025).
70. Bencivenga, D. *et al.* A cancer-associated CDKN1B mutation induces p27 phosphorylation on a novel residue: a new mechanism for tumor suppressor loss-of-function. *Mol Oncol* **15**, 915–941 (2021).
71. Wang, H., Guo, M., Wei, H. & Chen, Y. Targeting p53 pathways: mechanisms, structures and advances in therapy. *Sig Transduct Target Ther* **8**, 92 (2023).
72. Formaggio, N., Rubin, M. A. & Theurillat, J.-P. Loss and revival of androgen receptor signaling in advanced prostate cancer. *Oncogene* **40**, 1205–1216 (2021).
73. Butler, L. M. *et al.* Suppression of androgen receptor signaling in prostate cancer cells by an inhibitory receptor variant. *Mol Endocrinol* **20**, 1009–1024 (2006).

74. Tennakoon, J. B. *et al.* Androgens regulate prostate cancer cell growth via an AMPK-PGC-1α-mediated metabolic switch. *Oncogene* **33**, 5251–5261 (2014).
75. Siskin, M., Economides, M. P. & Wise, D. R. Cyclin-Dependent Kinase Inhibition in Prostate Cancer: Past, Present, and Future. *Cancers* **17**, (2025).
76. Dadwal, U. C., Chang, E. S. & Sankar, U. Androgen Receptor-CaMKK2 Axis in Prostate Cancer and Bone Microenvironment. *Frontiers in Endocrinology* **9**, 335 (2018).
77. Pulliam, T. L. *et al.* Regulation and role of CAMKK2 in prostate cancer. *Nat Rev Urol* **19**, 367–380 (2022).
78. Penfold, L. *et al.* CAMKK2 Promotes Prostate Cancer Independently of AMPK via Increased Lipogenesis. *Cancer Research* **78**, 6747–6761 (2018).
79. Thakur, S., Kumar, V., Das, R., Sharma, V. & Mehta, D. K. Biomarkers of Hepatic Toxicity: An Overview. *Current Therapeutic Research* **100**, 100737 (2024).
80. Martin, M. Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet.journal* **17**, 10–12 (2011).
81. Patro, R., Duggal, G., Love, M. I., Irizarry, R. A. & Kingsford, C. Salmon provides fast and bias-aware quantification of transcript expression. *Nat Methods* **14**, 417–419 (2017).
82. Soneson, C., Love, M. I. & Robinson, M. D. Differential analyses for RNA-seq: transcript-level estimates improve gene-level inferences. *F1000Res* **4**, 1521 (2015).
83. Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol* **15**, 550 (2014).
84. Ignatiadis, N., Klaus, B., Zaugg, J. B. & Huber, W. Data-driven hypothesis weighting increases detection power in genome-scale multiple testing. *Nat Methods* **13**, 577–580 (2016).
85. Stephens, M. False discovery rates: a new deal. *Biostatistics* **18**, 275–294 (2017).
86. Yu, G., Wang, L.-G., Han, Y. & He, Q.-Y. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS* **16**, 284–287 (2012).


PGC1α mRNA
PSMA-targeted PGC1α RNA NGs
Xenograft model of Prostate Cancer
Heart
Lungs
Spleen
Liver
Kidneys
Prostate
Tumour volume after 32 days (mm³)
1000
800
600
400
200
0
Untreated xenografts
PSMA_PGC1α_NGs

Research Highlights:

- First prostate-targeted redox-responsive nanogel for mRNA delivery
- PSMA targeting showed selective uptake in PSMA-positive cells
- NT-PGC1α mRNA enhanced mitochondrial abundance and respiration in PCa cells
- Treatment suppressed oncogenic programs and induced tumor suppressors
- NT-PGC1α mRNA decreased tumor growth in vivo by more than 70% with minimal toxicity